

Probability, Statistics & Data Analysis

Lecture 8: Non-Parametric Tests and Regression

**An approximate answer to the right
problem is worth a good deal more
than an exact answer to an
approximate problem**

John Tukey

Lecture 8: Non-parametric Tests and Regression

What we will cover

- Non-parametric tests
 - Sign test
 - Sign-rank test
 - Rank-sum test
- Categorical vs continuous analysis
- Correlation Coefficients
 - Pearson's and the Coefficient of Determination
 - Spearman's
- Linear Regression
- Confidence Intervals and NHST on regression parameters
- Confidence and Prediction Bands





Non-parametric Tests

When data doesn't fit a known distribution

Non-parametric tests for 1 or two samples

- Often, transformation will not be sufficient to 'enable' the use of parametric statistics
- Here we must turn to non-parametric statistics
- As a general rule, non-parametric tests are less powerful (less likely to correctly reject the null-hypothesis)

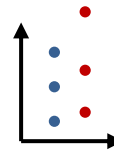
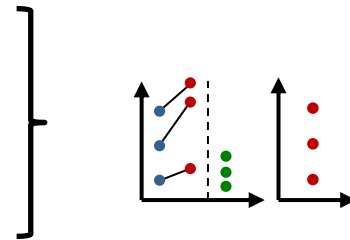
Non-parametric

- Sign test
- Sign-rank test
- Rank-sum test

Asymmetric



Symmetric



Parametric

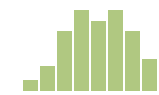
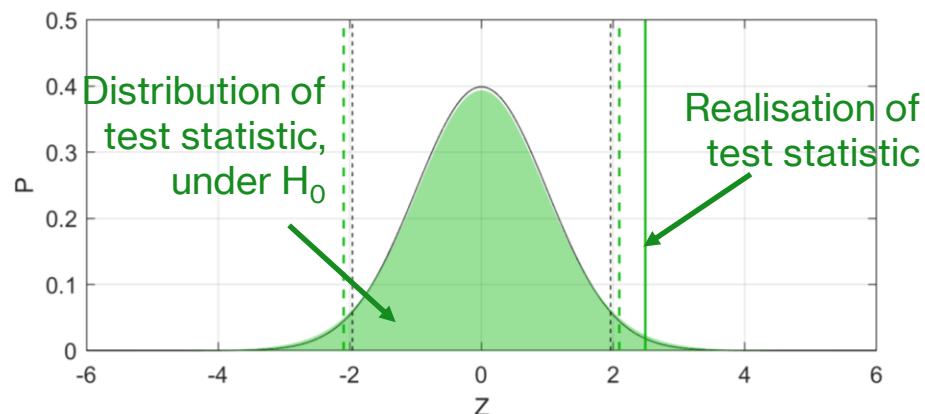
- Paired or 1 sample t -test
- Unpaired or 2 sample t -test

General Approach for Non-parametric Tests



For parametric tests

1. Calculate a test statistic using values
2. Compare to a parametric distribution of the outcome if the null hypothesis is true
3. If probability of observed outcome is 'small', reject H_0
4. Conclude something about population means



For non-parametric tests

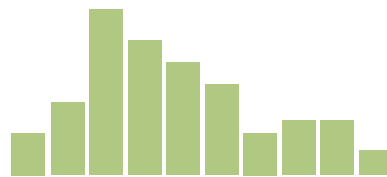
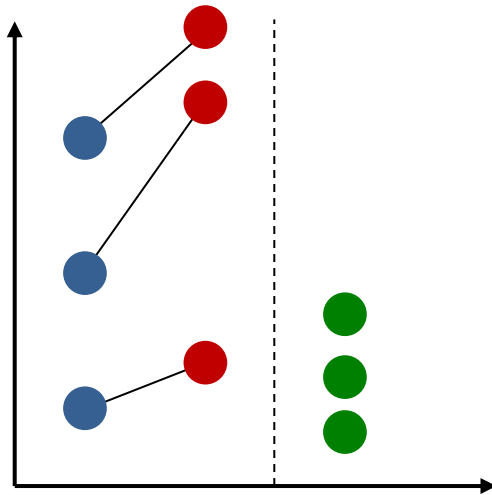
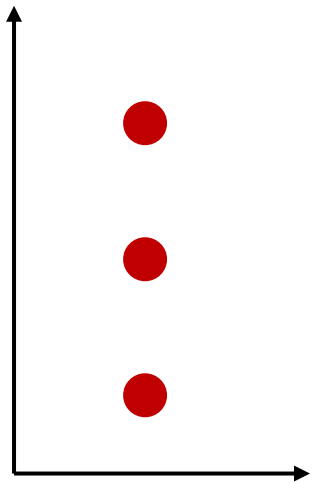
1. Calculate a test statistic using signs and/or ranks
2. Compare to probability of the outcome if the null hypothesis is true
3. If probability of observed outcome is 'small', reject H_0
4. Conclude something about population medians*

*with a few caveats

$$r_i = [1, 2, 3, 4, 5, 6, 7, 8, 9]$$

$$u = \sum_i^{n_1} r_{1,i} - \frac{n_1(n_1 + 1)}{2}$$

$$P(u \leq 0 | H_0) = \frac{1}{126} = 0.0079$$



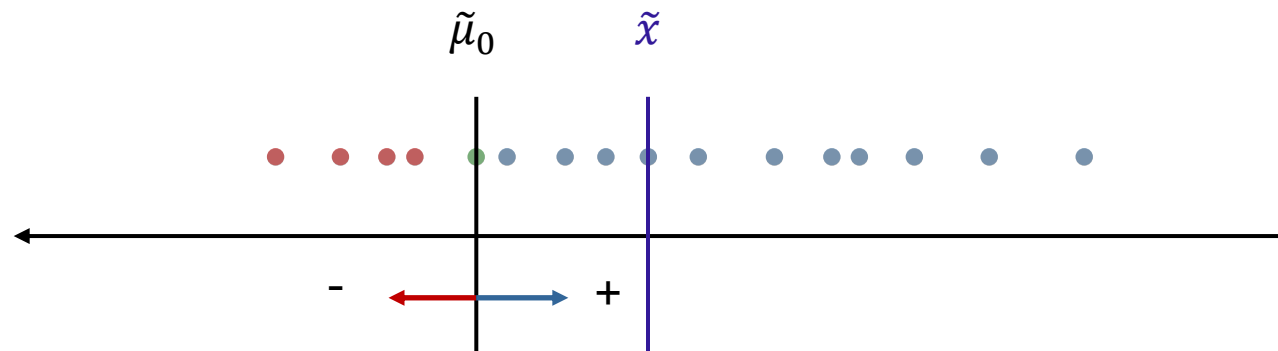
Sign Test

For asymmetric, non-normal, one-sample analyses

Sign Test

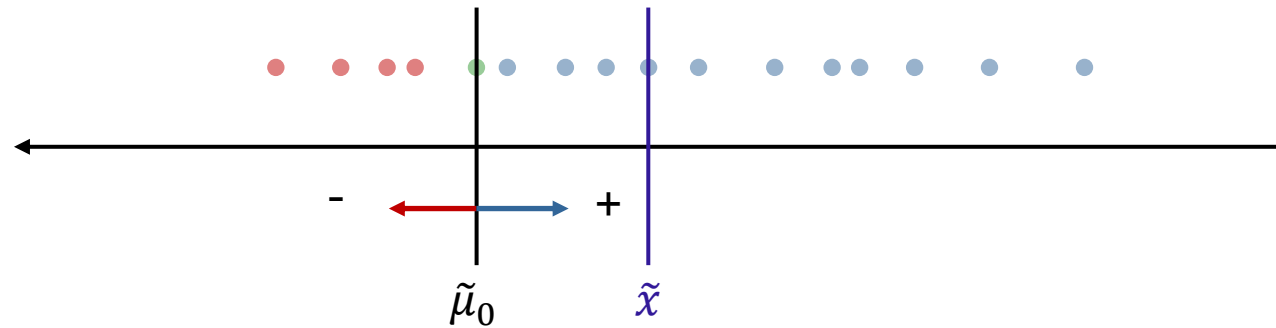
- A non-parametric alternative to one-sample t -test or paired t -test
 - Makes no assumptions about the shape of the distribution
 - Compares a sample median ($\tilde{x}$ or $\tilde{d}$) to a reference median value $\tilde{\mu}_0$
 - Only uses information about whether values are greater or lower than $\tilde{\mu}_0$, not by how much
 - Two-tailed hypotheses
 - H_0 : the population median $\tilde{\mu}$ from which the sample is taken is equal to $\tilde{\mu}_0$
 - H_1 : $\tilde{\mu} \neq \tilde{\mu}_0$
- In simple terms,
 - The test looks at whether each value is larger or smaller than $\tilde{\mu}_0$
 - Whether the **sign** of the difference is positive or negative
 - If H_0 is true, the probability that a given observation would be above (or below) $\tilde{\mu}_0$ is 0.5
 - If a large proportion of the sample is $> \tilde{\mu}_0$ or $< \tilde{\mu}_0$ then H_0 is unlikely

Note: median and mean are both 'averages'



signtest

Sign Test Algorithm



Step 1

For each x_i , let s_i be a realisation of a Bernoulli RV

$$s_i = \begin{cases} 0, & x_i - \tilde{\mu}_0 < 0 \\ 1, & x_i - \tilde{\mu}_0 > 0 \end{cases}$$

Discard $x_i = \tilde{\mu}_0$ (due to rounding)

Step 2

Define a binomial RV A according to

$$a = \sum_{i=1}^n s_i$$

Step 3

Use binomial distribution with $p = 0.5$ (null-hypothesis) to find $P(A = a) = b(a; n, 0.5)$

Probability of observed a if H_0 is true

Sign Test

Discard $x_i = \tilde{\mu}_0$

$$s_i = \begin{cases} 0, & x_i - \tilde{\mu}_0 < 0 \\ 1, & x_i - \tilde{\mu}_0 > 0 \end{cases}$$

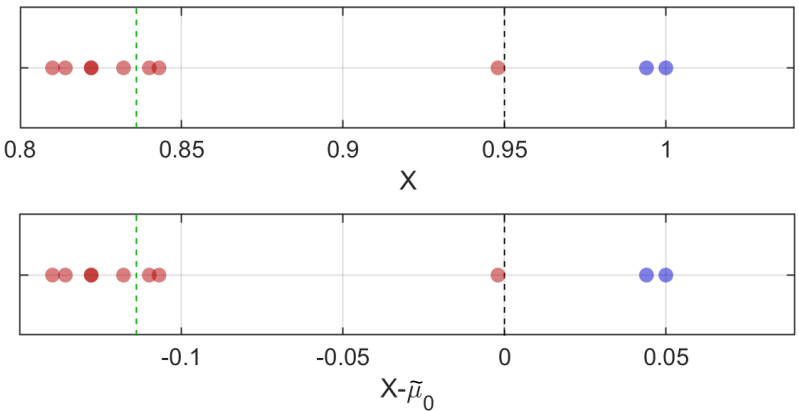
$$a = \sum_{i=1}^n s_i$$

$$P(A = a) = b(a; n, 0.5)$$

Cancer Cell Stiffness - Signs



- Malignant breast cancer cell mechanical stiffness as measured using atomic force microscopy in kPa
- We want to test the null hypothesis H_0 : the data are sampled from a distribution with median stiffness $\tilde{\mu}_0 = 0.95kPa$ (two-tailed H_0 - $\tilde{\mu} = \tilde{\mu}_0$)



x_i	0.840	0.810	0.832	0.814	0.822	1.000	0.822	0.948	0.994	0.843
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Step 1

$$s_i = \begin{cases} 0, & x_i - \tilde{\mu}_0 < 0 \\ 1, & x_i - \tilde{\mu}_0 > 0 \end{cases}$$

Discard $x_i = \tilde{\mu}_0$

$x_i - \tilde{\mu}_0$										
Signs										
s_i										

Sign Test

Discard $x_i = \tilde{\mu}_0$

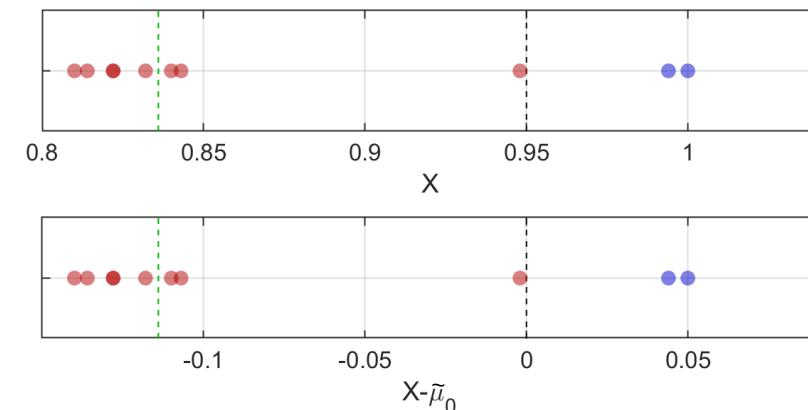
$$s_i = \begin{cases} 0, & x_i - \tilde{\mu}_0 < 0 \\ 1, & x_i - \tilde{\mu}_0 > 0 \end{cases}$$

$$a = \sum_{i=1}^n s_i$$

$$P(A = a) = b(a; n, 0.5)$$

How to get probability?

- If H_0 is true, how many data points would we expect to get that are positive?
- The answer is $n/2 = 5$ (from the definition of the median)
- Counting multiple Bernoulli RVs (the sign for each cell) yields a Binomial RV



Step 2 $a = \sum_{i=1}^n s_i = 2$ $b(x; n, p) = \binom{n}{x} p^x (1 - p)^{n-x}$

- If H_0 is true, what is the probability of getting this outcome or more extreme?
- What is the probability of a given observation being above the median, given H_0
- $p=0.5$

$$A \sim b(a; n, 0.5)$$

Sign Test

Discard $x_i = \tilde{\mu}_0$

$$s_i = \begin{cases} 0, & x_i - \tilde{\mu}_0 < 0 \\ 1, & x_i - \tilde{\mu}_0 > 0 \end{cases}$$

$$a = \sum_{i=1}^n s_i$$

$$P(A = a) = b(a; n, 0.5)$$

Sign Test - Probability

$$A \sim b(a; n, 0.5)$$

- What is the probability of getting this outcome or more extreme
- More extreme would be

$$H_1: \tilde{\mu} > \tilde{\mu}_0, p = P(A \geq a)$$

$$H_1: \tilde{\mu} < \tilde{\mu}_0, p = P(A \leq a)$$

$$H_1: \tilde{\mu} \neq \tilde{\mu}_0:$$

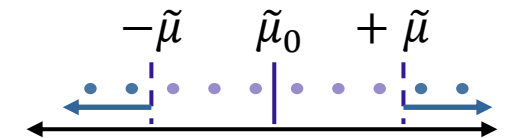
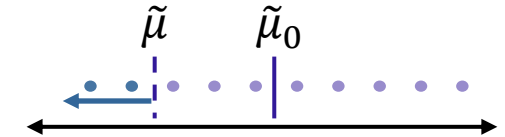
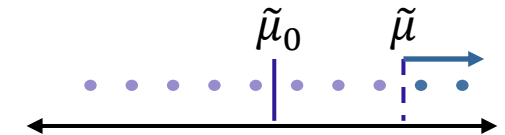
$$\circ \text{ for } x < n/2, p = 2P(A \leq a)$$

$$\circ \text{ for } x > n/2, p = 2P(A \geq a)$$

Upper-tailed

Lower-tailed

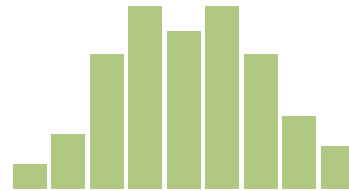
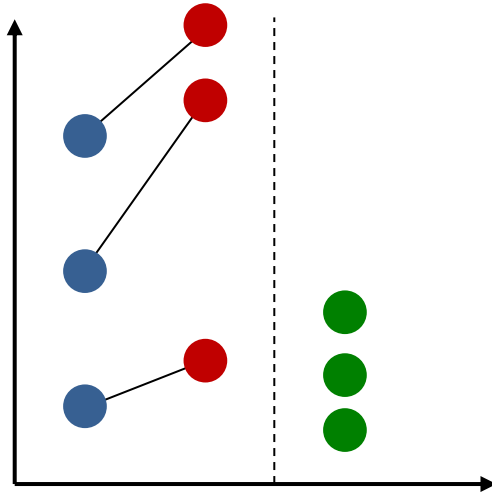
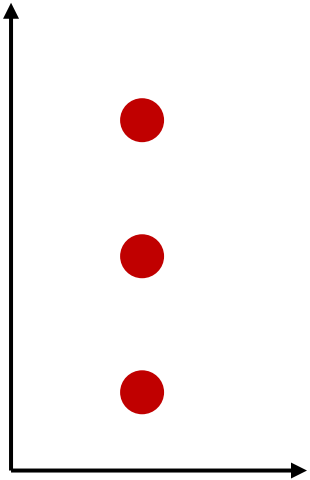
Two-tailed



Step 3 – for 2-tailed test

$$p(a) = \binom{n}{a} 0.5^a 0.5^{n-a}$$

$$\begin{aligned} \tilde{\mu} &= 0.836 \\ \tilde{\mu}_0 &= 0.95 \end{aligned}$$

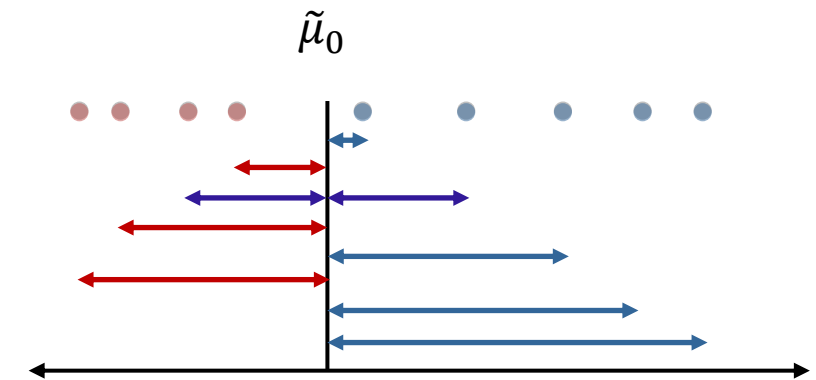


Sign-Rank Test

For symmetric, non-normal, one-sample analyses

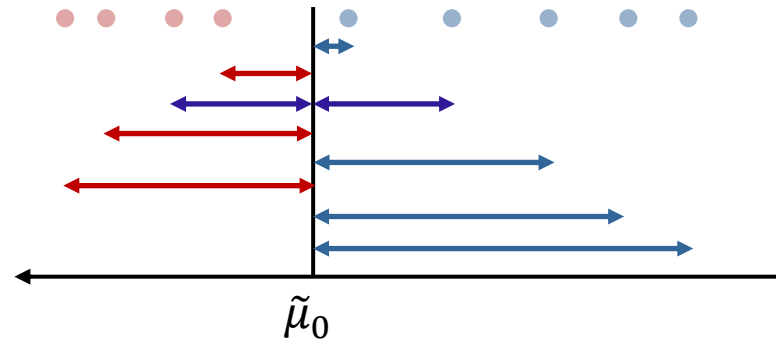
Sign-Rank Test

- A non-parametric alternative to one-sample t -test or paired t -test
 - Often ‘Wilcoxon sign-rank test’
 - Assumes data are from a symmetric distribution (i.e. not skewed)
 - The test looks at whether, and by how much, each value is larger or smaller than $\tilde{\mu}_0$
 - Two-tailed hypotheses
 - H_0 : the population median $\tilde{\mu}$ from which the sample is taken is equal to $\tilde{\mu}_0$
 - H_1 : $\tilde{\mu} \neq \tilde{\mu}_0$
- In simple terms,
 - The test looks at how much larger or smaller than $\tilde{\mu}_0$ each value is
 - Calculate ranks of distance from $\tilde{\mu}_0$
 - It can be thought of as a rank-weighted version of the sign test
 - If H_0 is true, positive ranks would approximately equal negative ranks
 - If large asymmetry in ranks, then H_0 is unlikely



signrank

Sign-Rank Test Algorithm



Step 1

For each x_i , find the product of the sign and rank of distance from $\tilde{\mu}_0$

$$r_i = \text{sign}(x_i - \tilde{\mu}_0) \times \text{rank}(|x_i - \tilde{\mu}_0|)$$

Discard $x_i = \tilde{\mu}_0$

For duplicate ranks, use average

Step 2

Define test statistic w

$$\begin{aligned} w^+ &= \sum_i^{n^+} r_i^+ \\ w^- &= -\sum_i^{n^-} r_i^- \\ w &= \min\{w^-, w^+\} \end{aligned}$$

Step 3

Compare w to w_{crit} from table

Reject if $w < w_{crit}$

- opposite to the parametric test statistics we've looked at!

Choice of w^+ , w^- or w depends on the tailedness of the test

Sign-Rank Test

Discard $x_i = \tilde{\mu}_0$
 $r_i = \text{sign}(x_i - \tilde{\mu}_0)$
 $\times \text{rank}(|x_i - \tilde{\mu}_0|)$

$$\begin{aligned} w^+ &= \sum_i^{n^+} r_i^+ \\ w^- &= -\sum_i^{n^-} r_i^- \\ w &= \min\{w^-, w^+\} \end{aligned}$$



Cancer Cell Stiffness - Sign-Ranks

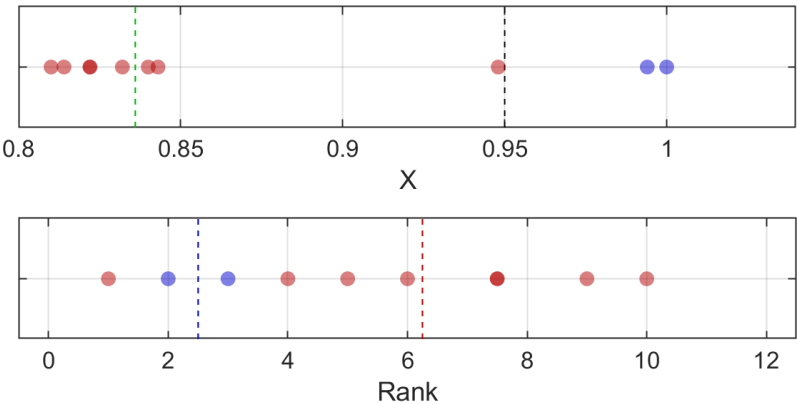
Step 1

x_i	0.840	0.810	0.832	0.814	0.822	1.000	0.822	0.948	0.994	0.843
$x_i - \tilde{\mu}_0$	-0.110	-0.140	-0.118	-0.136	-0.128	0.050	-0.128	-0.002	0.044	-0.107
Signs	-	-	-	-	-	+	-	-	+	-

r_i^+ = the positive ranks
 n^+ = # positive ranks

r_i^- = the negative ranks
 n^- = # negative ranks

Step 2



Sign-Rank Test

Discard $x_i = \tilde{\mu}_0$
 $r_i = \text{sign}(x_i - \tilde{\mu}_0)$
 $\times \text{rank}(|x_i - \tilde{\mu}_0|)$

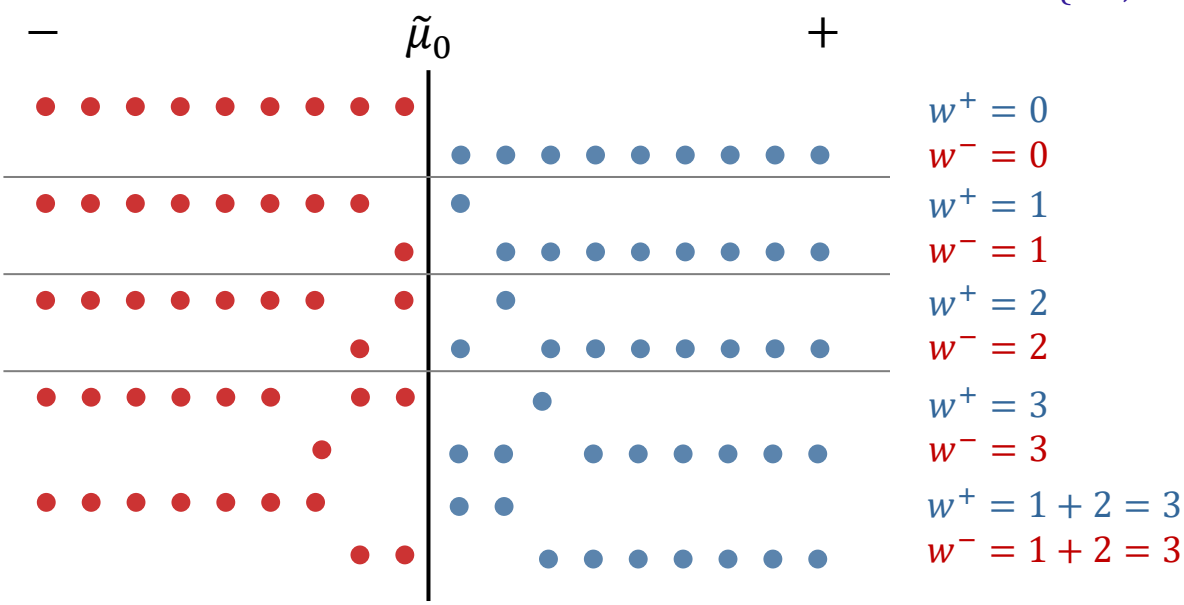
$$w^+ = \sum_i^{n^+} r_i^+$$
$$w^- = -\sum_i^{n^-} r_i^-$$
$$w = \min\{w^-, w^+\}$$

Understanding the w-statistic (2-tailed)

Ignoring tied ranks!

$$w^+ = \sum_i^{n^+} r_i^+$$
$$w^- = -\sum_i^{n^-} r_i^-$$
$$w = \min\{w^-, w^+\}$$

- For example, for n=2
 - Possible sign-rank (r_i) outcomes are
 - $r_i = \{-1, -2\}$ $w^+ = 0, w^- = 3, w = 0$
 - $r_i = \{1, -2\}$ $w^+ = 1, w^- = 2, w = 1$
 - $r_i = \{-1, 2\}$ $w^+ = 2, w^- = 1, w = 1$
 - $r_i = \{1, 2\}$ $w^+ = 3, w^- = 0, w = 0$
- 4 combinations in total
- Probability of e.g. getting $w = 0$ is 0.5
- In general,
 - 2^n possible combinations for the r_i values
 - E.g. n=8, there are $2^8=256$ ways in which the ranked-signs could be distributed
 - The probability of any given outcome is $\frac{1}{2^n} = 1/256 = 0.0039$



- $p(w \leq 0) = \frac{2}{256} = 0.0078$ [0] $p(A) = \frac{n(A)}{n}$
- $p(w \leq 1) = \frac{2+2}{256} = 0.0156$ [1]
- $p(w \leq 2) = \frac{4+2}{256} = 0.0234$ [2]
- $p(w \leq 3) = \frac{6+4}{256} = 0.0391$ [3], [1,2]
- $p(w \leq 4) = \frac{10+4}{256} = 0.0547$ [4], [1,3]
- $p(w \leq 5) = \frac{14+6}{256} = 0.0781$ [5], [1,4], [2,3]
- $p(w \leq 6) = \frac{20+6}{256} = 0.1016$ [6], [1,5], [2,4]

Our table chooses w_{crit} that minimises $|p - \alpha|$

Table 5: Critical values for the Sign-Rank Test			
n	One-Sided $\alpha = 0.01$ Two-Sided $\alpha = 0.02$	One-Sided $\alpha = 0.025$ Two-Sided $\alpha = 0.05$	One-Sided $\alpha = 0.05$ Two-Sided $\alpha = 0.1$
5			1
6		1	2
7	0	2	4
8	2	4	6
9	3	6	8

Cancer Cell Stiffness - w_{crit}

$$w^+ = 5$$

$$w^- = 50$$

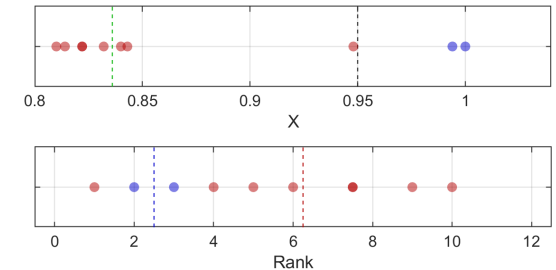
$$w = \min\{w^-, w^+\} = 5$$

n	One-Sided $\alpha = 0.01$	One-Sided $\alpha = 0.025$	One-Sided $\alpha = 0.05$
	Two-Sided $\alpha = 0.02$	Two-Sided $\alpha = 0.05$	Two-Sided $\alpha = 0.1$
5			1
6		1	2
7	0	2	4
8	2	4	6
9	3	6	8
10	5	8	11

- Critical value of w from the table is w_{crit}
- Let's use $\alpha = 0.05$
- For two-tailed, **reject** if $w \leq w_{crit}$

$$\tilde{x} = 0.836$$

$$\tilde{\mu}_0 = 0.95$$



- What about 'one-sided' ?
- For upper-tailed, **reject** if $w^- \leq w_{crit}$
- For lower-tailed, **reject** if $w^+ \leq w_{crit}$

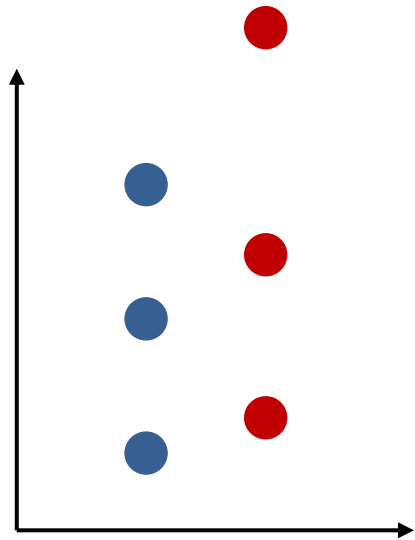
Sign-Rank Test

Discard $x_i = \tilde{\mu}_0$
 $r_i = \text{sign}(x_i - \tilde{\mu}_0)$
 $\times \text{rank}(|x_i - \tilde{\mu}_0|)$

$$w^+ = \sum_i^{n^+} r_i^+$$

$$w^- = -\sum_i^{n^-} r_i^-$$

$$w = \min\{w^-, w^+\}$$



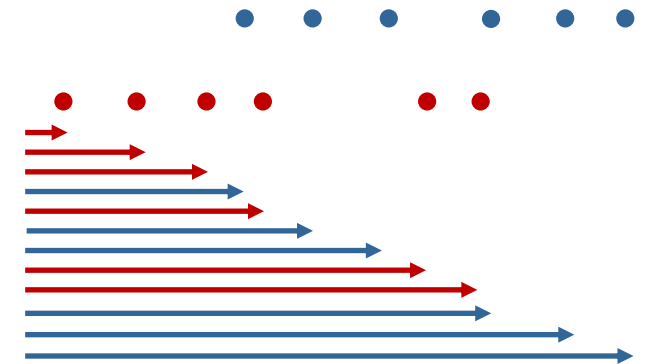
Rank-Sum test

For non-normal, two-sample analyses

Rank-Sum Test

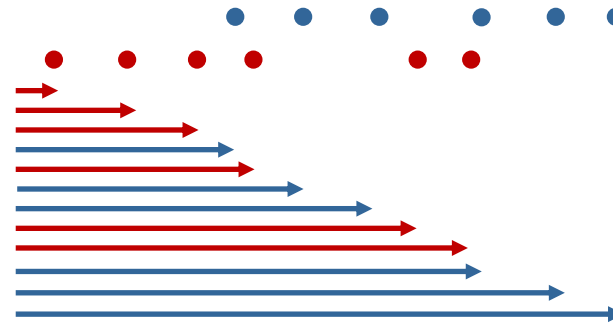


- A non-parametric alternative to two sample or unpaired Student's t -test
 - Also known as Wilcoxon Rank-Sum Test and Mann-Whitney U-test
 - MATLAB calls the function ranksum
 - We use a U-statistic
 - Compares two sample medians to each other
 - Two-tailed
 - H_0 : the underlying population medians are equal, $\tilde{\mu}_1 = \tilde{\mu}_2$
 - H_1 : $\tilde{\mu}_1 \neq \tilde{\mu}_2$
 - Under the assumption that
 - Independent and from similar underlying distributions (similar to homoscedastic)
 - Otherwise, we can still use the test, but technically the null hypothesis becomes that the mean ranks are the same
- In simple terms
 - Gather both samples together and assign a rank to each data point
 - Calculate the sum of the ranks and then a U-statistic for each group
 - If H_0 is true, ranks for each group would be approximately equal
 - If one group has larger ranks, then H_0 is unlikely
- If you want to compare to some $\tilde{\mu}_0$ (not common)
 - E.g. $H_0: \tilde{\mu}_1 - \tilde{\mu}_2 \neq \tilde{\mu}_0$ but not in reference sheet for clarity



ranksum

Rank-Sum Test Algorithm



Step 1

Choose samples 1 and 2 such that $n_1 \leq n_2$
Combine data sets and find the rank of each data point

$$r_i = \text{rank}(\{x_1, x_2\})$$

For duplicate ranks, use average

Calculate w-statistics

$$w_1 = \sum_i^{n_1} r_{1,i}, \quad w_2 = \sum_i^{n_2} r_{2,i}$$

Step 2

Calculate test statistic u

$$u_i = w_i - \frac{n_i(n_i+1)}{2}, \quad u = \min\{u_1, u_2\}$$

For this course, if asked for a test statistic, use the u-statistic (which may need to be calculated)

Step 3

Compare u to u_{crit} from table

Reject if $u < u_{crit}$

- opposite to the parametric test statistics we've looked at!

Choice of u_1 , u_2 or u depends on the tailedness of the test

Rank-Sum Test

$$r_i = \text{rank}\{x_1, x_2\},$$

$$n_1 \leq n_2$$

$$w_1 = \sum_i^{n_1} r_{1,i}$$

$$w_2 = \sum_i^{n_2} r_{2,i}$$

$$u_i = w_i - \frac{n_i(n_i + 1)}{2}$$

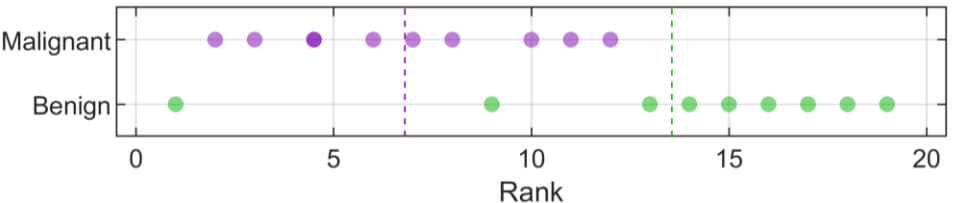
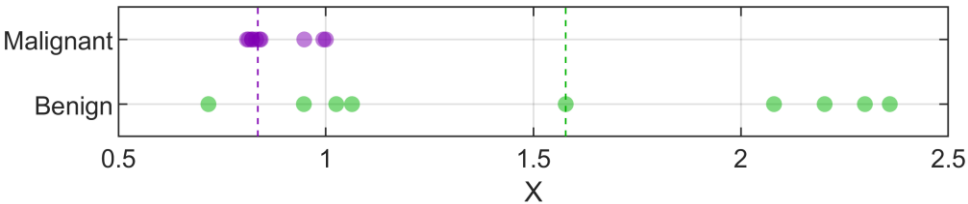
$$u = \min\{u_1, u_2\}$$

Calculating sum ranks



- We now compare malignant to benign cells
- Test null hypothesis: $\tilde{\mu} = \tilde{\mu}_0$, with $\alpha = 0.05$

Malignant (x_2)	0.840	0.810	0.832	0.814	0.822	1.000	0.822	0.948	0.994	0.843
Benign (x_1)	0.717	1.025	2.202	2.359	1.578	0.947	2.299	1.063	2.080	-



Step 1 Group both samples, rank and correct averages

Rank, r_i	1	2	3	4.5	4.5	6	7	8	9	10
Value	0.717	0.810	0.814	0.822	0.822	0.832	0.840	0.843	0.947	0.948
Rank, r_i	11	12	13	14	15	16	17	18	19	
Value	0.994	1.000	1.025	1.063	1.578	2.080	2.202	2.299	2.359	

$\tilde{x}_1 = 1.58$
 $\tilde{x}_2 = 0.84$

Calculate the sum ranks for each group (W)

Rank-Sum Test

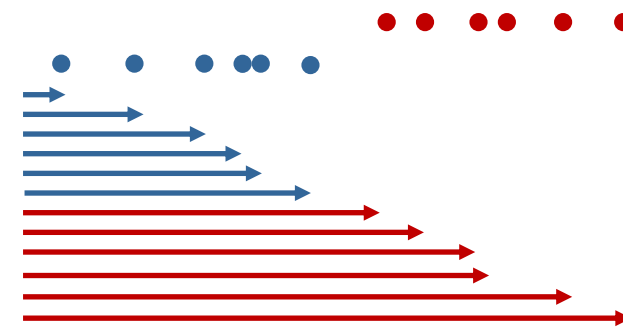
$$r_i = \text{rank}\{x_1, x_2\},$$
$$n_1 \leq n_2$$
$$w_1 = \sum_i^{n_1} r_{1,i}$$
$$w_2 = \sum_i^{n_2} r_{2,i}$$

$$u_i = w_i - \frac{n_i(n_i + 1)}{2}$$
$$u = \min\{u_1, u_2\}$$

FYI - Why convert from W to U?

- Consider n_1 samples with $w_1 = \sum_{i=1}^{n_1} r_{1,i}$
- Consider the extreme case where all $r_1 < r_2$
 - $w_1 = \sum_{i=1}^{n_1} i = 1 + 2 + 3 + \dots + n_1 = n_1(n_1 + 1)/2$
- What would w_2 equal?
 - $w_2 = \sum_{i=1}^{n_2} (n_1 + i) = \sum_{i=1}^{n_2} n_1 + \sum_{i=1}^{n_2} i = n_1 n_2 + n_2(n_2 + 1)/2$
- If we transform our w stats to u stats according to

$$u_i = w_i - \frac{n_i(n_i + 1)}{2}$$
 - $u_1 = 0$
 - $u_2 = n_1 n_2$
- Hence, $0 < U < n_1 n_2$
- This is more intuitive and makes tables easier to work with



n	$\sum_{i=1}^n i$	$n + 1$	$n(n + 1)$	$n(n + 1)/2$
1	1	2	2	1
2	3	3	6	3
3	6	4	12	6
4	10	5	20	10

Note, MATLAB's `ranksum` outputs the w statistic, not the u statistic
 For this course, if asked for a test statistic, use the u -statistic (which may need to be calculated)

Understanding the u-statistic (2-tailed)

Ignoring tied ranks!

- There are $(n_1 + n_2)$ ranks, and n_1 choices of ranks for sample 1
- When H_0 is true, any combination is as likely as any other combination
 - E.g. $n_1 = 4, n_2 = 5$
 - There are $\binom{9}{4} = \frac{9!}{4!5!} = \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 126$ combinations – note $\binom{9}{4} = \binom{9}{5}$
- $p(u \leq 0) = \frac{2}{126} = 0.0159$
- $p(u \leq 1) = \frac{2+2}{126} = 0.0317$
- $p(u \leq 2) = \frac{4+4}{256} = 0.0635$

This table chooses u_{crit} such that $p < \alpha$

	u_{crit}	
	n_2	
n_1	4	5
1		
2		
3		0
4	0	1
5		2

$$u_i = w_i - \frac{n_i(n_i + 1)}{2}$$
$$u = \min\{u_1, u_2\}$$

$$\frac{n_1(n_1 + 1)}{2} = \frac{4(4 + 1)}{2} = 10$$

$$\frac{n_2(n_2 + 1)}{2} = \frac{5(5 + 1)}{2} = 15$$

For purple larger

● ● ● ● ● ● ● ● ● ●

$w_1 = 10, w_2 = 35, u_1 = 0, u_2 = 20, u = 0$

● ● ● ● ● ● ● ● ● ●

$w_1 = 11, w_2 = 34, u_1 = 1, u_2 = 19, u = 1$

● ● ● ● ● ● ● ● ● ●

$w_1 = 12, w_2 = 33, u_1 = 2, u_2 = 18, u = 2$

● ● ● ● ● ● ● ● ● ●

$w_1 = 12, w_2 = 33, u_1 = 2, u_2 = 18, u = 2$

● ● ● ● ● ● ● ● ● ●

$w_1 = 13, w_2 = 32, u_1 = 3, u_2 = 17, u = 3$

... and so on

For green larger

● ● ● ● ● ● ● ● ● ●

$w_1 = 30, w_2 = 15, u_1 = 20, u_2 = 0, u = 0$

● ● ● ● ● ● ● ● ● ●

$w_1 = 29, w_2 = 16, u_1 = 19, u_2 = 1, u = 1$

● ● ● ● ● ● ● ● ● ●

$w_1 = 28, w_2 = 17, u_1 = 18, u_2 = 2, u = 2$

● ● ● ● ● ● ● ● ● ●

$w_1 = 28, w_2 = 17, u_1 = 18, u_2 = 2, u = 2$

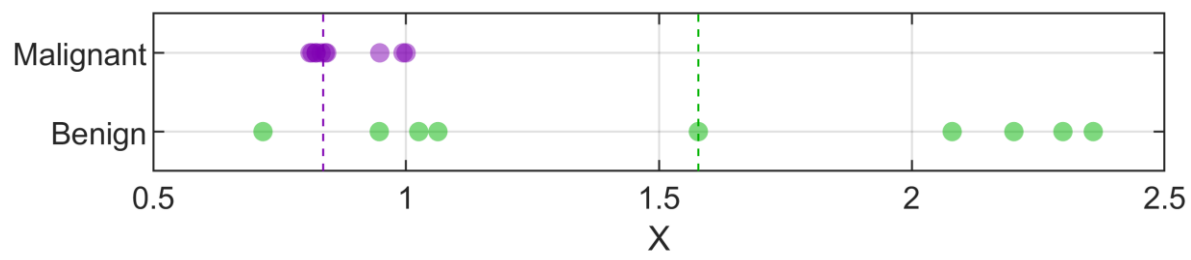
● ● ● ● ● ● ● ● ● ●

$w_1 = 27, w_2 = 18, u_1 = 17, u_2 = 3, u = 3$

... and so on



Cancer Cell Stiffness - u_{crit}



$\tilde{x}_2 = 0.84$
 $\tilde{x}_1 = 1.58$

Table for two-tailed $\alpha = 0.05$

n_1	n_2									
	4	5	6	7	8	9	10	11	12	13
1										
2					0	0	0	0	1	1
3		0	1	1	2	2	3	3	4	4
4	0	1	2	3	4	4	5	6	7	8
5		2	3	5	6	7	8	9	11	12
6			5	6	8	10	11	13	14	16
7				8	10	12	14	16	18	20
8					13	15	17	19	22	24
9						17	20	23	26	28
10							23	26	29	33

Step 3

- For two-tailed, ($H_1: \tilde{\mu}_2 \neq \tilde{\mu}_1$), reject if $u \leq u_{crit}$
- What about ‘one-sided’ (relative to $\tilde{\mu}_1$)?
- For upper-tailed ($H_1: \tilde{\mu}_2 < \tilde{\mu}_1$), reject if $u_2 \leq u_{crit}$
- For lower-tailed ($H_1: \tilde{\mu}_2 > \tilde{\mu}_1$), reject if $u_1 \leq u_{crit}$

Table for one-tailed $\alpha = 0.05$

n_1	n_2									
	3	4	5	6	7	8	9	10	11	12
1										
2			0	0	0	1	1	1	1	2
3	0	0	1	2	2	3	4	4	5	5
4		1	2	3	4	5	6	7	8	9
5			4	5	6	8	9	11	12	13
6				7	8	10	12	14	16	17
7					11	13	15	17	19	21
8						15	18	20	23	26
9							21	24	27	30
10								27	31	34



Summary of 1 and 2 sample non-parametric-tests

for reference and understanding

Non-parametric tests

```
[p,h,STATS] = signtest(x,med0,'method','exact')
```

```
p=0.1094
```

```
h=0 % 'conclude' h0
```

```
stats=
```

```
zval: NaN % approximate z-value for large samples - ignore this
```

```
sign: 2 % the test statistic, a
```

Stops it approximating with normal distribution for large samples

```
[p,h,STATS] = signrank(x,med0,'method','exact')
```

```
p=0.0195
```

```
h=1 % 'conclude' h1
```

```
stats=
```

```
signedrank: 5 % w-statistic (always gives w+)
```

Stops it approximating with normal distribution for large samples

```
[p,h,STATS] = ranksum(x1,x2,'method','exact')
```

```
p=0.0073
```

```
h=1 % 'conclude' h1
```

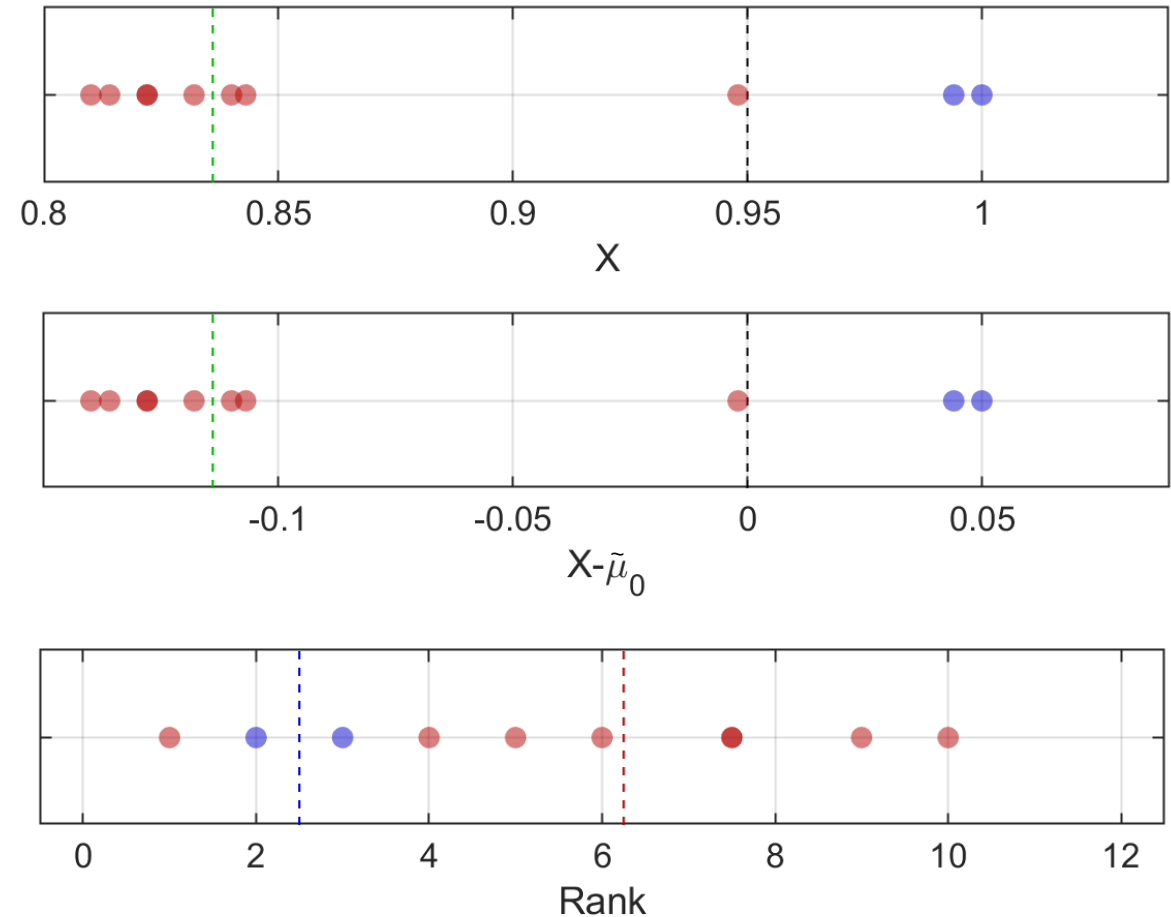
```
stats=
```

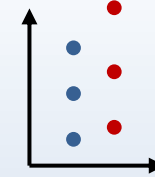
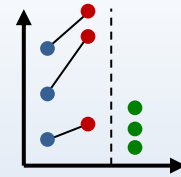
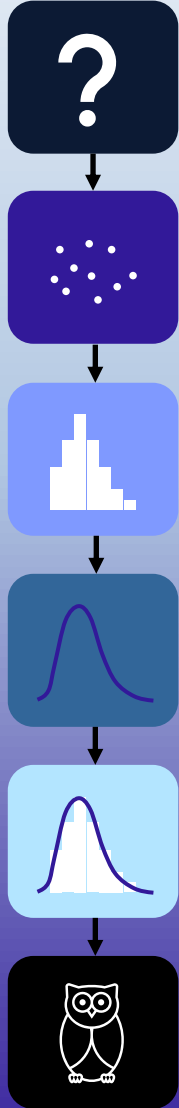
```
ranksum: 122 % w-statistic of the first value (here x1) not u  
statistic, which you can calculate
```

Stops it approximating with normal distribution for large samples

Sign test vs Sign-rank test

- We concluded different things
 - Sign test, $p=0.11$
 - Sign-rank test, $p=0.02$
- As the sign test doesn't include magnitude
 - the shape of the distribution around the **median** doesn't matter
- The sign-rank test takes into account the magnitude
 - so the large data points that make the distribution asymmetric have an undue influence on the outcomes
- Distribution of data is clearly asymmetric, so trust the sign test result
- Sign-rank test has
 - Less chance of Type II error for symmetric data
 - More chance of Type I error for asymmetric data
- Note both can be used on a single sample median ($\tilde{x}$) or for paired data on the median of the differences ($\tilde{d}$)





Sign test	Sign-rank	Rank-sum test
How does the ...		
population median $\tilde{\mu}_X$ or population median difference $\tilde{\mu}_D$		difference between the medians of the two populations, $\tilde{\mu}_1$ and $\tilde{\mu}_2$
... compare to some known value $\tilde{\mu}_0$?		(or add $\tilde{\mu}_0$ to x_1 or x_2)
Capture $i = 1:n$ observations, x_i or $i = 1:n$ paired observations, $x_{1,i}$ & $x_{2,i}$ to yield $d_i = x_{2,i} - x_{1,i}$		Capture $i = 1:n_1$ & $j = 1:n_2$ unpaired observations, $x_{1,i}$ & $x_{2,j}$
Visualisation of x_i or $x_{1,i}$ & $x_{2,i}$ and/or d_i		Visualisation of $x_{1,i}$ & $x_{2,j}$
x_i or d_i ... are sampled from a.		$x_{1,i}$ and $x_{2,j}$ are sampled from
an asymmetric distribution	a symmetric distribution	similarly distributed populations
$s_i = \begin{cases} 0, & x_i - \tilde{\mu}_0 < 0 \\ 1, & x_i - \tilde{\mu}_0 > 0 \end{cases}$ $a = \sum_{i=1}^n s_i$ $P(A = a) = b(a; n, 0.5)$	$r_i = \text{sign}(x_i - \tilde{\mu}_0) \times \text{rank}(x_i - \tilde{\mu}_0)$ $w^+ = \sum_i^{n^+} r_i^+$ $w^- = -\sum_i^{n^-} r_i^-$ $w = \min\{w^-, w^+\}$	$r_i = \text{rank}\{x_1, x_2\}$ $w_1 = \sum_i^{n_1} r_{1,i}, w_2 = \sum_i^{n_2} r_{2,i}$ $u_i = w_i - \frac{n_i(n_i+1)}{2}$ $u = \min\{u_1, u_2\}$
$H_0: \tilde{\mu}_X$ or $\tilde{\mu}_D$ or $\tilde{\mu}_2 - \tilde{\mu}_1$ is not 'different to'/'greater than'/'less than' $\tilde{\mu}_0$ (two-/upper-/lower-tailed) Is observed test statistic likely if H_0 is true, for a given α ? {yes/no}		
We found a median difference of $\tilde{x}$ or $\tilde{d}$ or $\tilde{y}$ that {was/was not} statistically different from $\tilde{\mu}_0$		

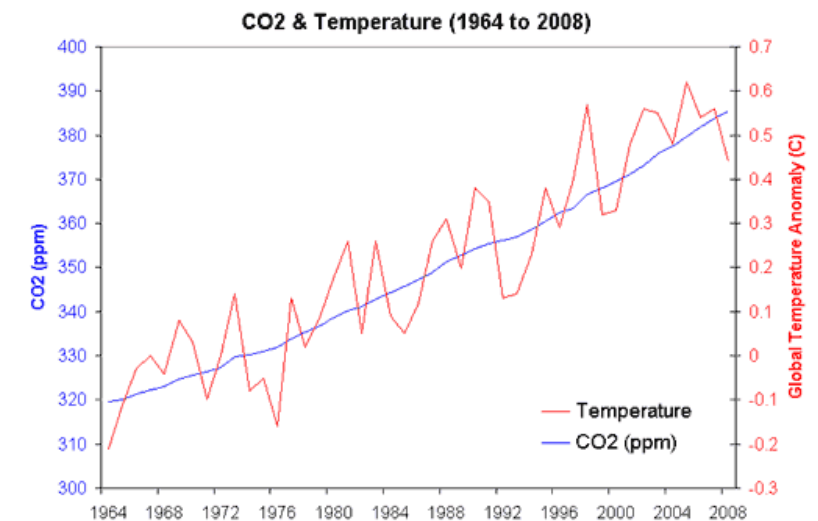
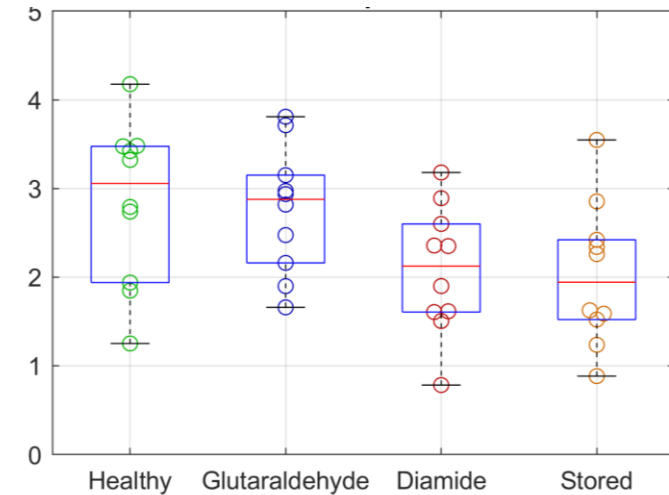


Continuous vs Categorical

When we're not just comparing outcomes for different groups

Categorical vs Continuous

- Thus far, we have considered testing hypotheses regarding 1 or two group means or medians
 - Treated vs control
 - Wild-type vs Knock-out
- Can also compare more than 2 means/medians
 - Different manufacturing techniques
 - Different treatment of RBCs
 - Covered next week
- These are categorical variables
- What if we are interested in different levels of a continuous variable
 - Does height affect weight?
 - Does atmospheric CO₂ affect global temperatures
 - In both cases, the answer is yes, but so do other things....



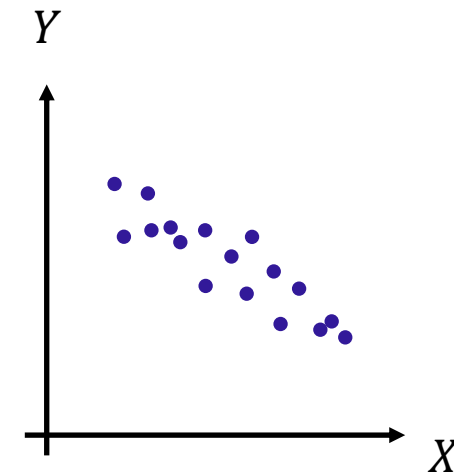
Acid Rain and Tree Dieback



- Forest Growth Research in Canada in 1995
- X is the pH of the soil
- Y is the % of dieback of sugar maple trees
- The question we want to ask
 - Does soil pH cause dieback?
- The question we can ask with statistics
 - Is there a correlation between dieback and soil pH?
- If so, then how much?



X	3.3	8.2	4.2	7.5	...
Y	18.3	6.3	13.1	10.4	...





Pearson's Correlation Coefficient

for linearly related data

Sample Correlation Coefficient, R

- We have already seen this for population data when we were looking at joint probability distributions
- There, we knew things about the population, including probabilities
- For statistics, we infer them

- We don't know σ or μ ,
 - but can calculate S and $\bar{X}$
- We don't know probabilities for all possible outcomes
 - but have a finite sample of realisations x_i
- We can't calculate ρ ,
 - but we can use r as a point estimator: $R = \hat{\rho}$

$$\text{cov}(X, Y) = \sum_x \sum_y (x - E(X))(y - E(Y))p(x, y)$$

$$\rho_{X,Y} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Introduce S_{xx} notation for sum squares

$$\sigma_X^2 \longrightarrow S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\sigma_Y^2 \longrightarrow S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$\text{cov}(X, Y) \longrightarrow S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Relationship to Population Correlation Coefficient



$$\sigma_X^2 \longrightarrow$$

$$\sigma_Y^2 \longrightarrow$$

$$\text{cov}(X, Y) \longrightarrow$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

$$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2$$

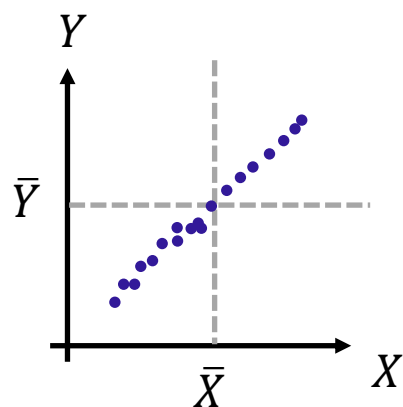
$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$\rho_{X,Y} = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y}$$

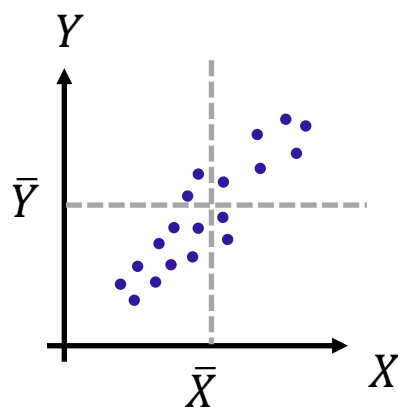


$$R(x, y) = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$$

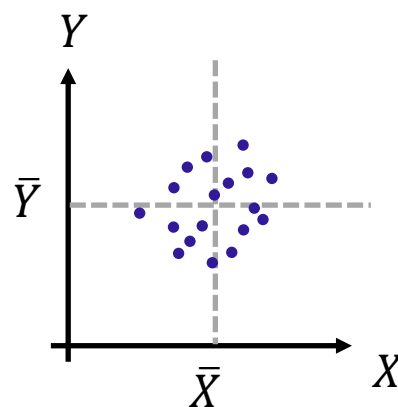
$$r(x, y) \approx 1$$



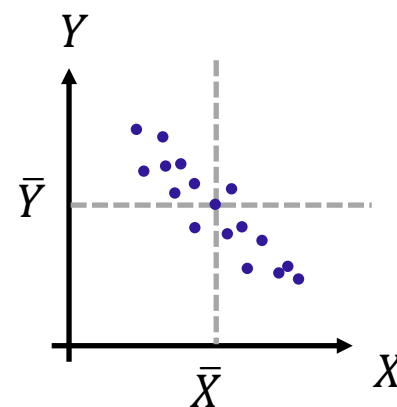
$$0 \leq r(x, y) \leq 1$$



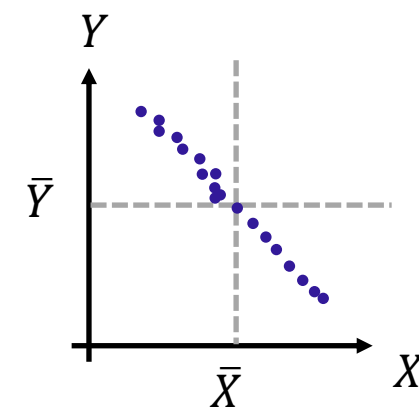
$$r(x, y) \approx 0$$



$$-1 \leq r(x, y) \leq 0$$

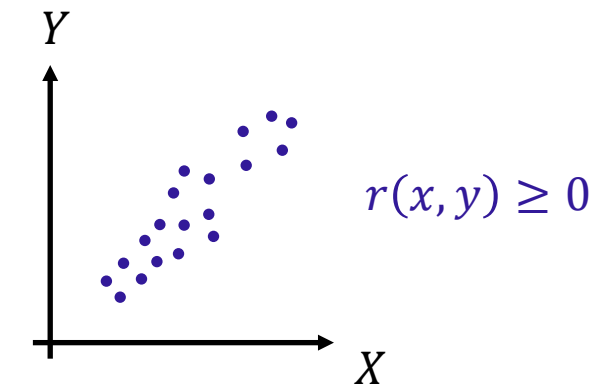
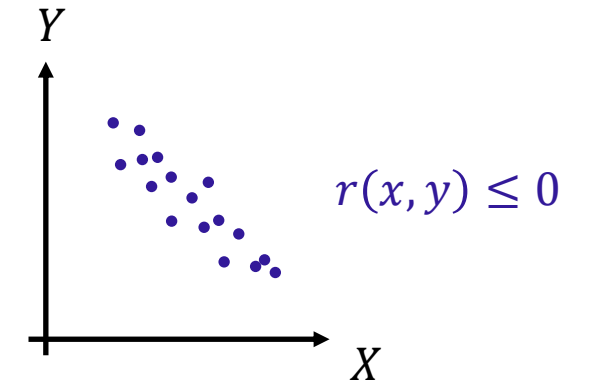
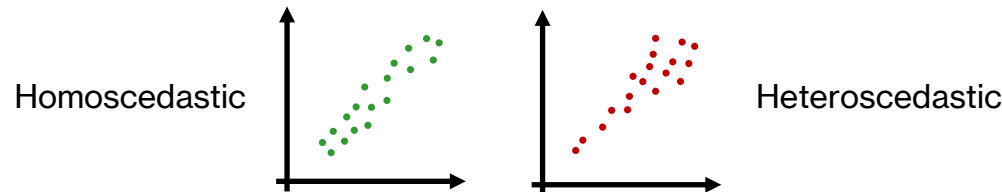


$$r(x, y) \approx -1$$



Important points about R

- $-1 \leq R \leq 1$
 - Negative R indicates that Y decreases as X increases
 - Positive R indicates that Y increases as X increases
 - When concluding, refer to real-world X and Y !
- $R(x, y) = R(y, x)$
- R is independent of the units used
 - Would be pretty useless if not!
- Also known as the Pearson Product-Moment Correlation Coefficient
- Assumptions
 - Continuous variables
 - Linear relationship between X and Y
 - Homoscedastic residuals



Hypothesis Testing R

- R is a point estimator for the population: $R = \hat{\rho}$
- 'What is the probability that we got our observed value of R if the null hypothesis is true?'
 - H_0 : there is no correlation, $R = 0$
 - H_1 : there is a correlation, $R \neq 0$
- We can develop a test statistic for R
 - Requires transforming R to be ~normally distributed
- $v = n - 2$ degrees of freedom
 - Uses S_{xy} which has $n - 2$ DOF
- Can calculate p -value as usual with ρ in place of μ (and their sample equivalents)
- Important
 - p tells us if ' r is statistically different from 0'
 - r tells us if the dependence is large or small
 - neither tell us if the effect (slope) is practically significant

$$R(x, y) = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

$$T = \frac{R\sqrt{n-2}}{\sqrt{1-R^2}}$$

$$\rho > 0$$



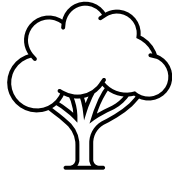
$$\rho < 0$$



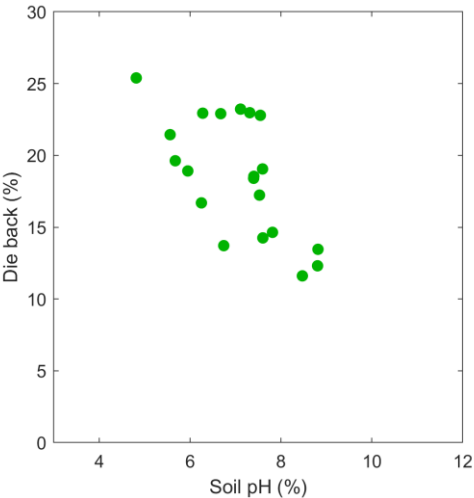
$$\rho \neq 0$$



Acid Rain and Tree Dieback



X is the pH of the soil
Y is the % of dieback of sugar maple trees



$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = 21.98$$

$n = 20$

$$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = 329.6$$

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = -55.52$$

		α						
1-tailed		0.1	0.05	0.025	0.01	0.005	0.001	0.0005
2-tailed		0.2	0.1	0.05	0.02	0.01	0.002	0.001
De	10	1.357	1.740	2.120	2.585	2.921	3.000	4.013
	17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
	18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
	19	1.328	1.729	2.093	2.539	2.861	3.579	3.883

- Negatively correlated
- Decreasing pH (acidification) correlates with increasing dieback, $p < 0.002$
- Our sample correlation coefficient, r , is statistically different from zero

Coefficient of Determination

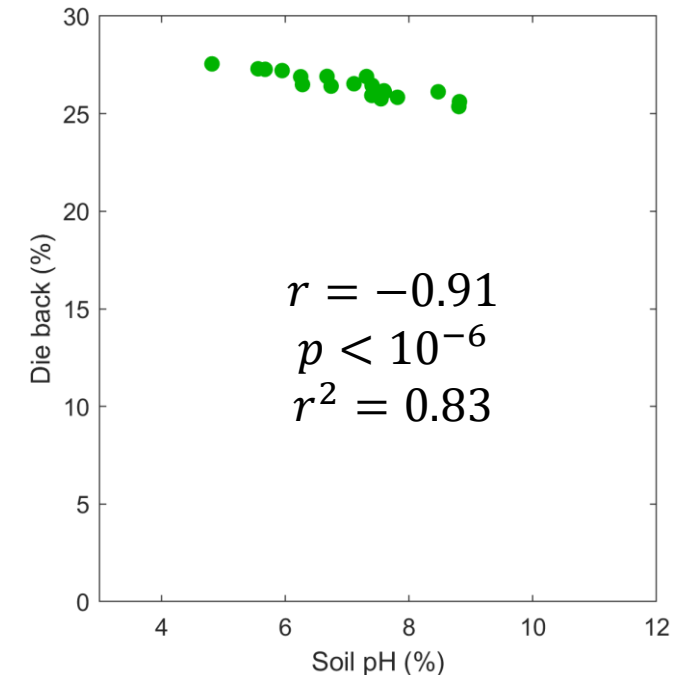


- How to interpret $r = -0.65$

Coefficient of determination = R^2

Individual realisations of r^2 are regularly written as R^2 even though technically inconsistent with RV notation

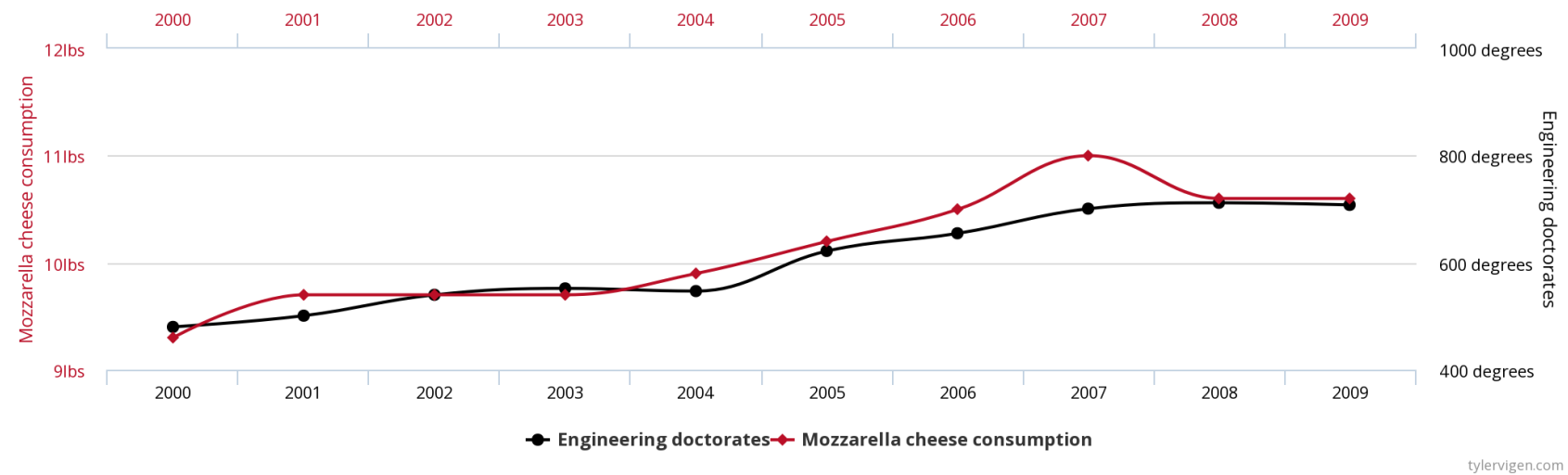
- Typically interpreted as
 - How much the variability in Y is explained by X (and vice versa)
 - How much the variability in Y is accounted for by X (and vice versa)
- More accurately (but less intuitively) interpreted as
 - How much of the variability is shared between X and Y
- Here $r^2 = 0.43$
 - 43% of the variability is shared between soil pH and % dieback
- r^2 tells us the strength of the dependence
 - Does not tell us the relevance of the effect
 - Should not be used to quantify the 'goodness-of-fit' for an obvious correlation** (see bonus rant video)
- High r^2 does not indicate causality
 - It could be that a certain type of beetle acidifies the soil and eats the tree roots
 - Well-designed experiments can get at causality
 - E.g. grow trees in a lab, change the soil pH



Correlation is not causation

Spurious Correlations

Correlation is not causation

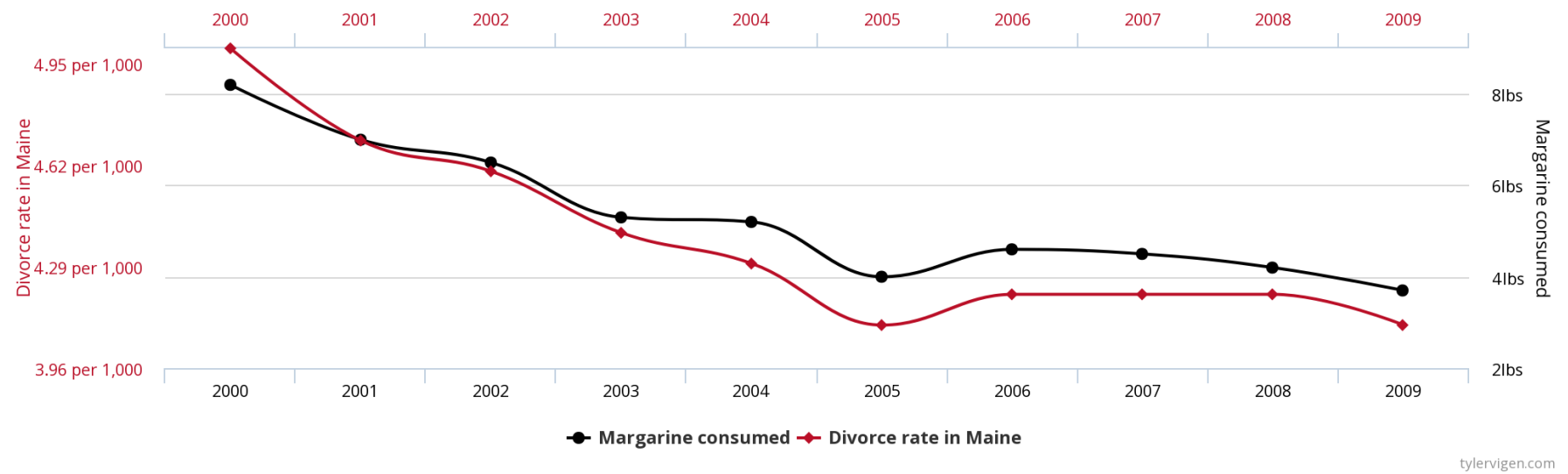


Mozzarella cheese consumption vs civil engineering doctorates awarded

$r^2 = 0.92$

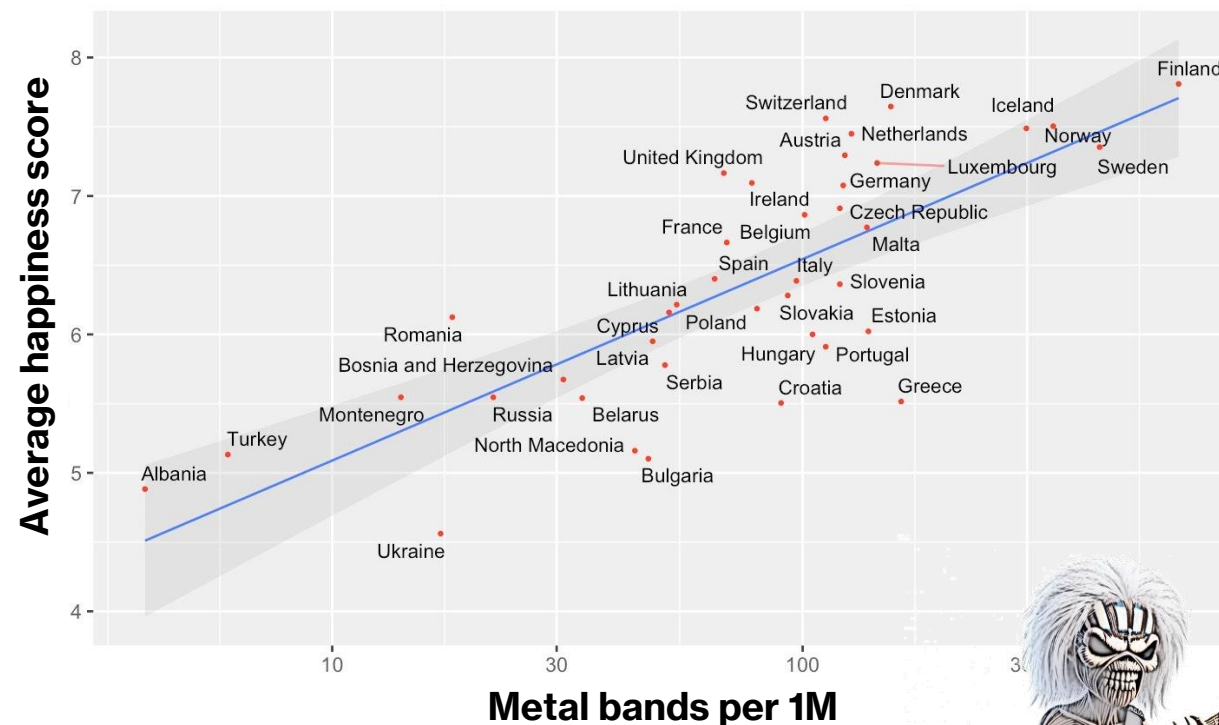
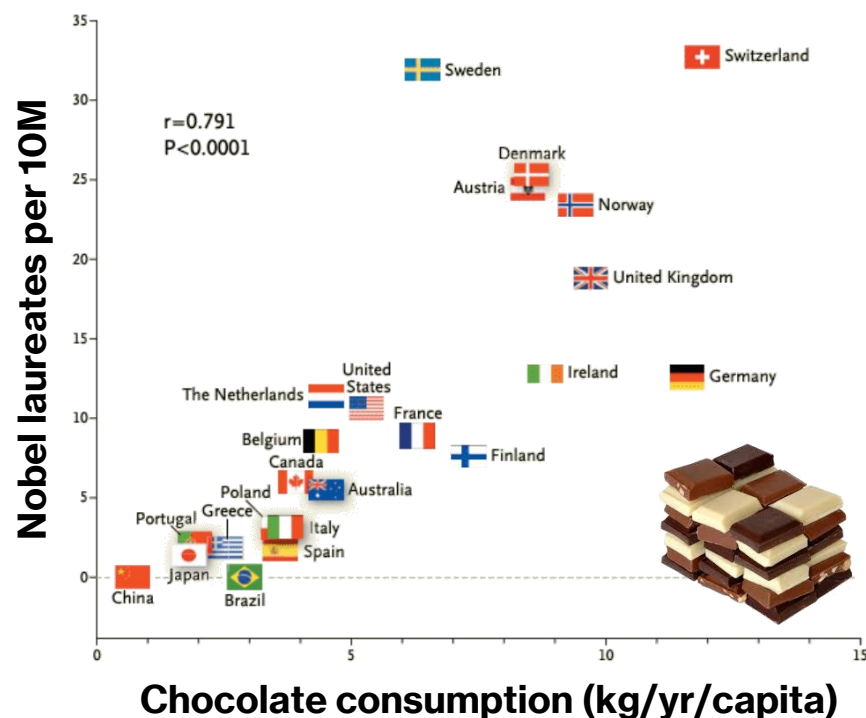
Divorce rate in Maine vs margarine consumed

$r^2 = 0.99$



www.tylervigen.com/spurious-correlation

Definitely not Spurious Correlations



- Chocolate makes you smart
- Metal makes you happy
- If there is causation, there probably is correlation (if you have enough data)....

<https://designer.microsoft.com/image-creator>

<https://p.migdal.pl/blog/2023/01/metal-bands-happiness-correlation/>

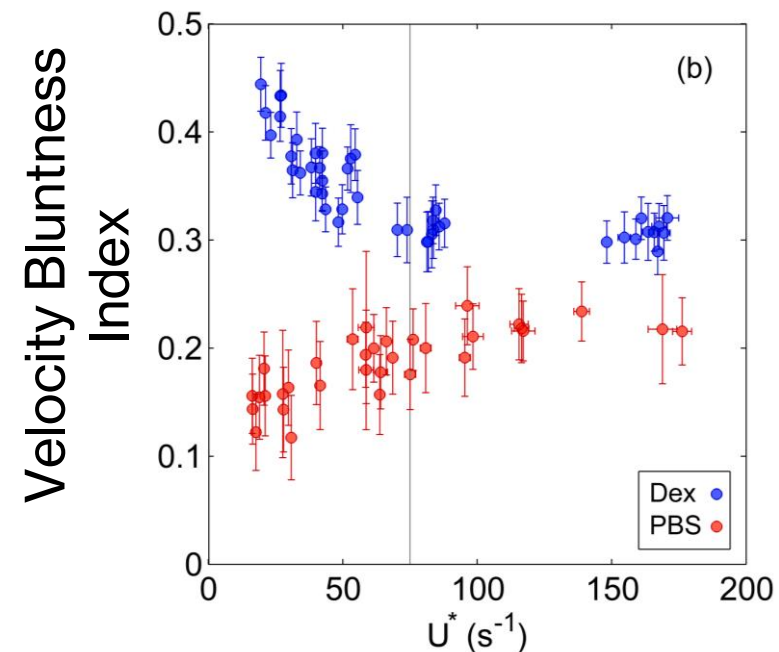
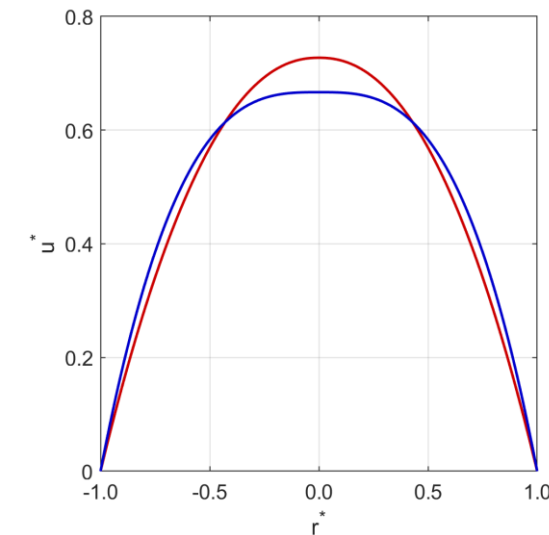


Spearman's Rank Correlation Coefficient

Non-parametric correlation

Spearman's Rank Correlation Coefficient, ρ_S

- Recall that Pearson's Correlation Coefficient assumed a linear relationship between X and Y
- What do we do when this is not the case?
- Example: 'bluntness indices' describing the shape of velocity profiles of blood in a microchannel as average velocity changes
- Note ρ_S is my terminology to avoid confusion with $\rho_{X,Y}$ (population correlation coefficient)



$$\rho_S = -0.84$$

$$\rho_S = 0.82$$

[Sherwood et al 2014](#)

Spearman's Correlation

- Rank each group independently
- Tied scores are given an averaged value
- Then calculate Pearson's R on the ranks

$$\begin{aligned}r_{x,i} &= \text{rank}\{x\} \\ r_{y,i} &= \text{rank}\{y\} \\ \rho_S &= R(r_x, r_y)\end{aligned}$$

- If there are no ties, it reduces to

$$\rho_S = 1 - \frac{6 \sum (r_{x,i} - r_{y,i})^2}{n(n^2 - 1)}$$

- But why learn two formulae?

Spearman's Example

```
x=10:10:180;
y=round(x.^(-0.6)+0.3+randn(size(x))*0.02,2)
```

← Second argument of
round sets number of
decimal places

```
[rho,p]=corr(x',y','type','Spearman')
```

```
rho=-0.7016
```

```
p=0.0012
```

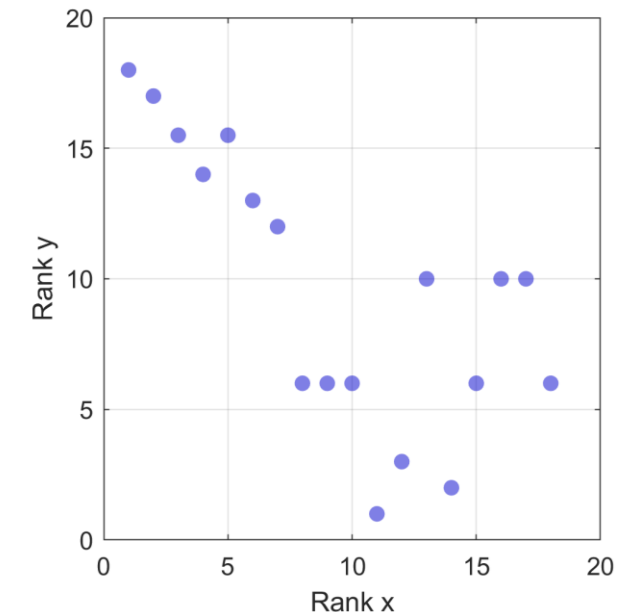
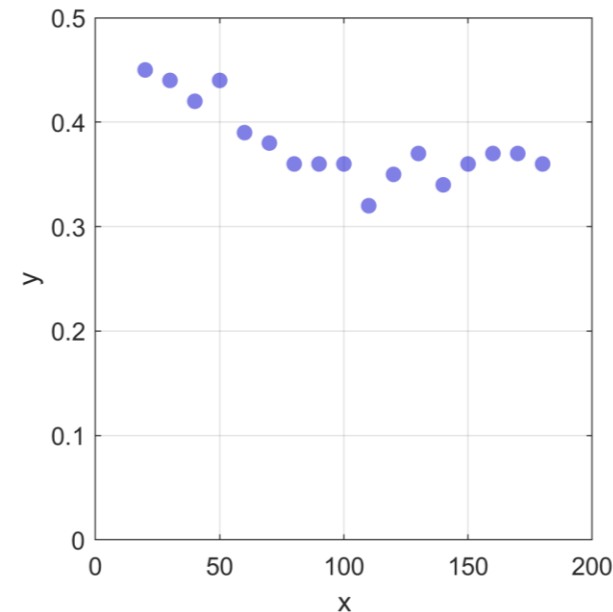
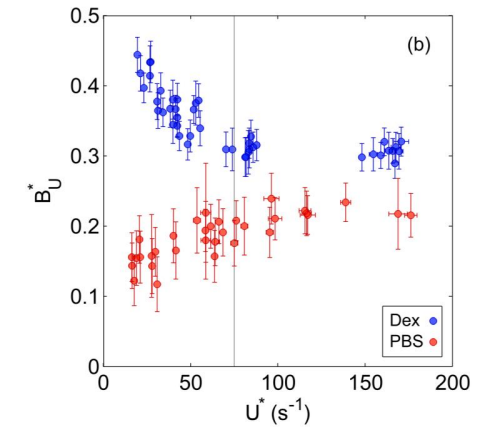
```
Rx=tiedrank(x);
Ry=tiedrank(y);
```

tiedrank – MATLAB function
that automates the ranking
and deals with ties

```
[rho,p]=corr(Rx',Ry','type','Pearson')
```

```
rho=-0.7016
```

```
p=0.0012
```

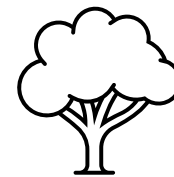




Linear Regression

Beyond $y=mx+c$

Two models for the data



- Let us consider a general formulation

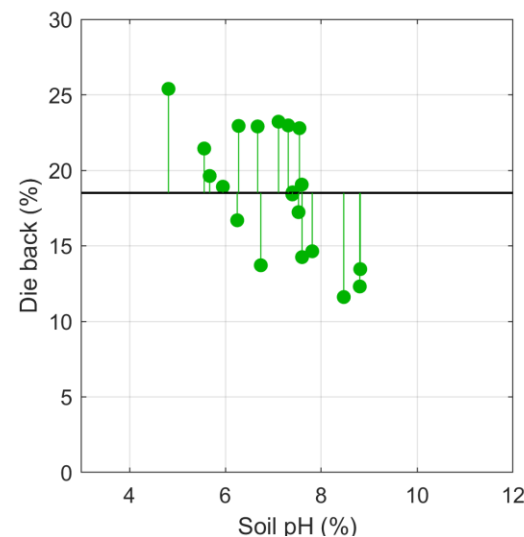
$$outcome_i = (model) + error_i$$

- For example, we could use the mean as the model
- The model could be written as
$$y_i = (c) + \varepsilon_i$$
 - For each observation i , the vertical line indicates the error ε_i
- For linear regression, we add a dependence of y on x

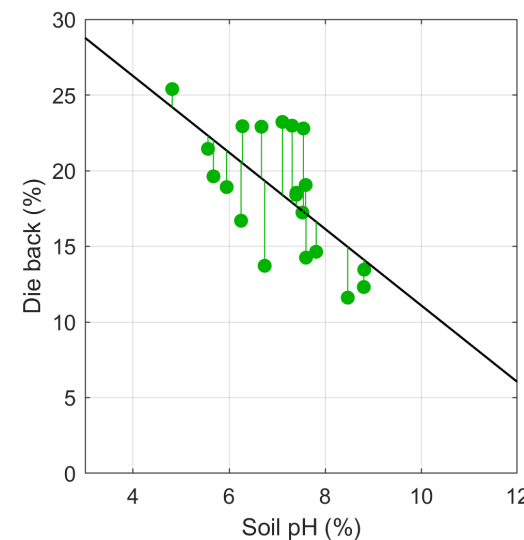
$$y_i = (c + mx_i) + \varepsilon_i$$

or

$$y_i = (b_0 + b_1x_i) + \varepsilon_i$$



H_0 : mean



H_1 : best fit line

Regression



- Writing Y as a statistic, rather than a single observation

Model $Y = b_0 + b_1X + \varepsilon$

- ε is a normally distributed random variable with $E(\varepsilon) = 0$ and $V(\varepsilon) = \sigma_\varepsilon^2$
- For each data point ε_i is the 'residual'

$$\varepsilon_i = y_i - (b_0 + b_1x_i)$$

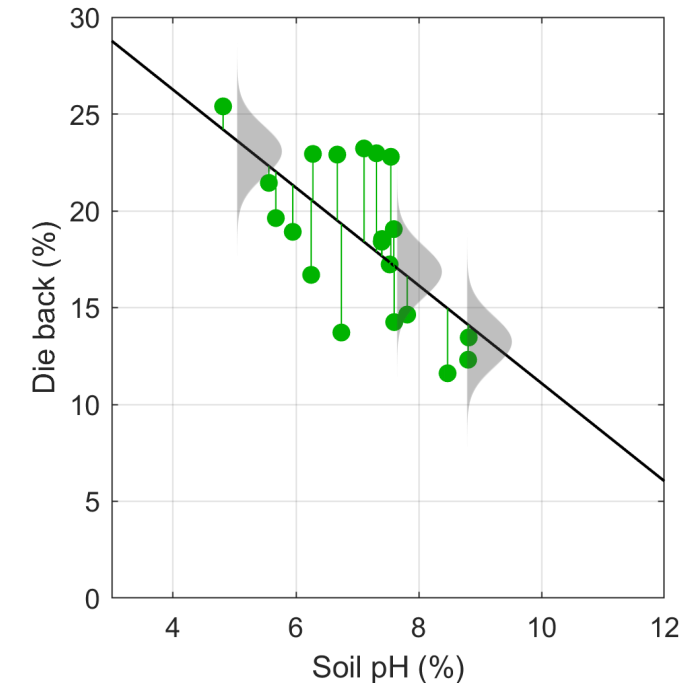
- The method of least squares finds the line that minimises the sum of the squared residuals, $f(b_0, b_1)$

$$f(b_0, b_1) = \sum_{i=1}^n \varepsilon_i^2$$

- Point estimates of b_0 and b_1 , denoted by $\widehat{b}_0$ and $\widehat{b}_1$ minimise $f(b_0, b_1)$ to get the regression line

Best-fit $Y = \widehat{b}_0 + \widehat{b}_1X$

$$y_i = (b_0 + b_1x_i) + \varepsilon_i$$



Note: i) each y_i must be independent, ii) sometimes people would average multiple y_i values at a given x_i prior to regression. If the multiple y_i were

- independent: you should use all data points rather than 'throw away data'
- dependent: then you should average first, so that the values you analyse with regression are independent.

FYI: Minimising Residuals – b_0

- Minimum value of a function occurs where the derivative is equal to zero (in this case, we don't require the second derivative)

$$\left. \frac{\partial f(b_0, b_1)}{\partial b_0} \right|_{\widehat{b}_0, \widehat{b}_1} = 0$$

$$f(b_0, b_1) = \sum_{i=1}^n [y_i - (b_0 + b_1 x_i)]^2$$

$$\frac{\partial f(b_0, b_1)}{\partial b_0} = -2 \sum_{i=1}^n [y_i - (b_0 + b_1 x_i)]$$

$$\frac{\partial f(b_0, b_1)}{\partial b_0} = -2 \left(\sum_{i=1}^n y_i - \sum_{i=1}^n b_0 - \sum_{i=1}^n b_1 x_i \right)$$

$$\frac{\partial f(b_0, b_1)}{\partial b_0} = -2(n\bar{y} - nb_0 - nb_1\bar{x})$$

$$\frac{\partial f(b_0, b_1)}{\partial b_0} = -2(n\bar{y} - nb_0 - nb_1\bar{x})$$

$$\frac{\partial f(b_0, b_1)}{\partial b_0} = \bar{y} - b_0 - b_1\bar{x}$$

$$\left. \frac{\partial f(b_0, b_1)}{\partial b_0} \right|_{\widehat{b}_0, \widehat{b}_1} = \bar{y} - \widehat{b}_0 - \widehat{b}_1\bar{x} = 0$$

$$\widehat{b}_0 = \bar{y} - \widehat{b}_1\bar{x}$$

FYI: Minimising Residuals – b_1

- Minimum value of a function occurs where the derivative is equal to zero (in this case, we don't require the second derivative)

$$\left. \frac{\partial f(b_0, b_1)}{\partial b_1} \right|_{\widehat{b}_0, \widehat{b}_1} = 0$$

$$f(b_0, b_1) = \sum_{i=1}^n [y_i - (b_0 + b_1 x_i)]^2$$

$$\left. \frac{\partial f(b_0, b_1)}{\partial b_1} \right|_{\widehat{b}_0, \widehat{b}_1} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) - \widehat{b}_1 \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\left. \frac{\partial f(b_0, b_1)}{\partial b_1} \right|_{\widehat{b}_0, \widehat{b}_1} = -2 \sum_{i=1}^n x_i [y_i - (\widehat{b}_0 + \widehat{b}_1 x_i)]$$

$$\widehat{b}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Using $\left\{ \begin{array}{l} \widehat{b}_0 = \bar{y} - \widehat{b}_1 \bar{x} \\ \sum_{i=1}^n x_i y_i - n \bar{x} \bar{y} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \end{array} \right.$

$$\widehat{b}_1 = \frac{S_{xy}}{S_{xx}}$$

Calculating the Variance

- So we now have our estimates of the 'best' fit line

$$Y = \widehat{b}_0 + \widehat{b}_1 X$$

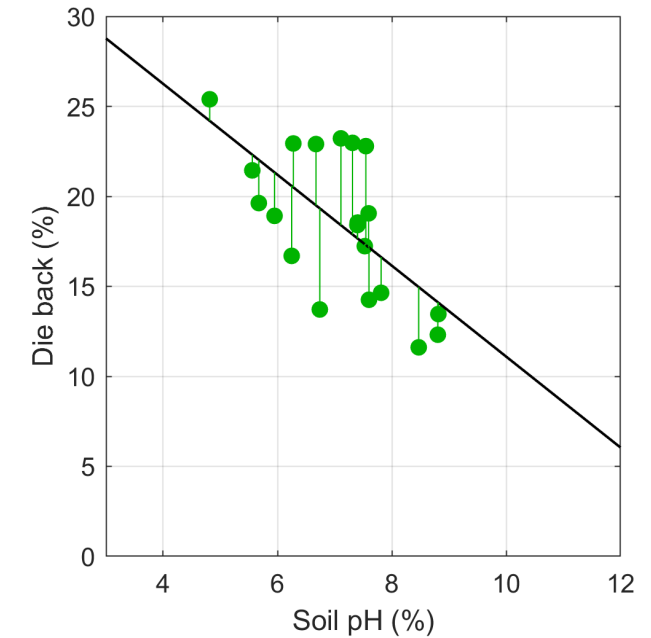
- But how good is it? We need to characterise ε
- It has mean of zero and variance σ_ε^2
 - Using the model to estimate $\widehat{y}_i$ for a given x_i

$$\widehat{y}_i = (\widehat{b}_0 + \widehat{b}_1 x_i)$$

- σ_ε^2 is the variance of the residuals $(y_i - \widehat{y}_i)$
- Our best estimator for σ_ε^2 is thus

$$\widehat{\sigma_\varepsilon^2} = s_\varepsilon^2 = \frac{SS_R}{\nu_R} = \frac{\sum_{i=1}^n (y_i - \widehat{y}_i)^2}{n - 2}$$

- SS_R is the sum of squares of the residuals (from our model, H_1)



$$SS_R = \sum_{i=1}^n (y_i - \widehat{y}_i)^2$$

$$\nu_R = n - 2$$

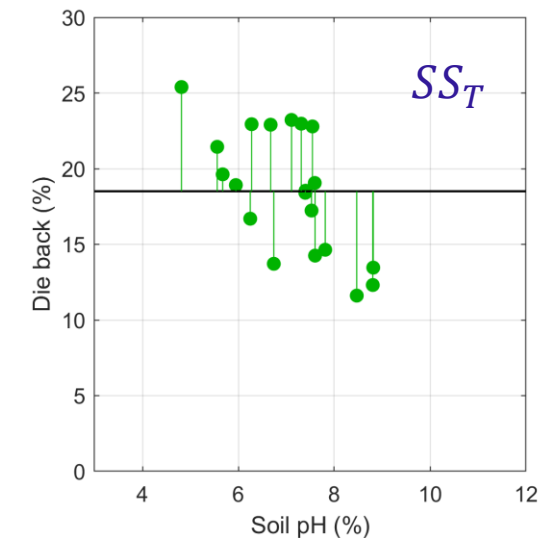
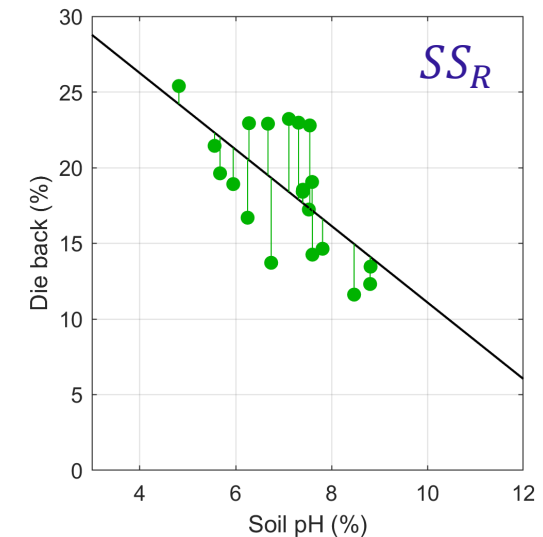
Now we introduced SS_R , let's revisit R^2

- SS_R is the sum of squares of the residuals from our model, H_1
- SS_T is the sum of squares of the residuals from the null model, i.e. the mean (H_0)

$$SS_T = \sum_{i=1}^n (y_i - \bar{y})^2 = S_{yy}$$

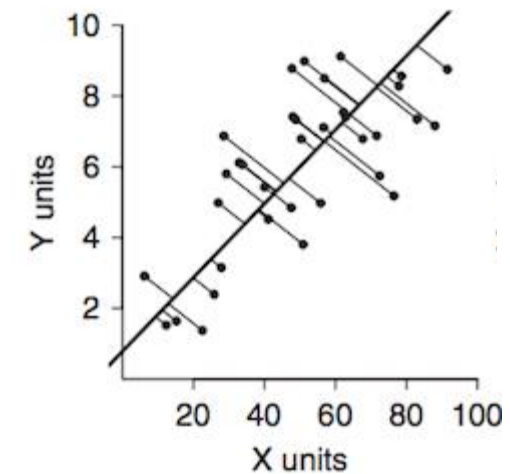
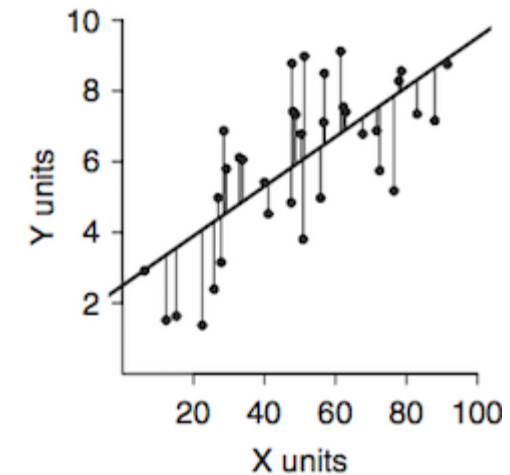
$$R^2 = 1 - \frac{SS_R}{SS_T}$$

- If $\frac{SS_R}{SS_T} = 1$ the linear fit and mean are equally good
 - $r^2 = 0$
- If $\frac{SS_R}{SS_T} = 0$ the linear fit is perfect
 - $r^2 = 1$



Comments on linear regression approaches

- Linear least squares regression only considers deviations from the model in the y axis
 - E.g. Dieback %
- What if there is uncertainty in the x-axis?
 - E.g. uncertainty in the measurement of soil pH
- Ordinary least squares (OLS) minimises y-residuals
- Reduced major axis (RMA) regression minimises residuals in both axes
- Can also account for known uncertainties on each y-value using weighted regression
 - Each y-residual gets divided by its variance in the minimisation
- Or on both variables
 - See York and Evensen 2004 (on Blackboard)



https://biostat2.uni.lu/lectures/lecture09_regression.html#12

Biostatistic Design and Analysis Using R. (Wiley-Blackwell, 2011) Murray Logan



How confident are we?

Confidence intervals

Confidence Intervals on Parameters



- We have established that we should never report a sample mean without giving confidence intervals
 - If we took 100 samples, 95 of the CIs would contain μ
- Likewise, we should not report regression coefficients without confidence intervals
- Indeed, the value of $\hat{y}_i = (\hat{b}_0 + \hat{b}_1 x_i)$ is the mean Y for a given X
- Each estimate of $\hat{y}_i$ has a confidence interval

$$\hat{y}_i \pm t_{1-\frac{\alpha}{2}, \nu_R} s_{\hat{y}}$$

- First, let's look at the estimates of the slope and intercept

Confidence Interval on the Slope

- The confidence interval on $\widehat{b}_1$ will follow the general form

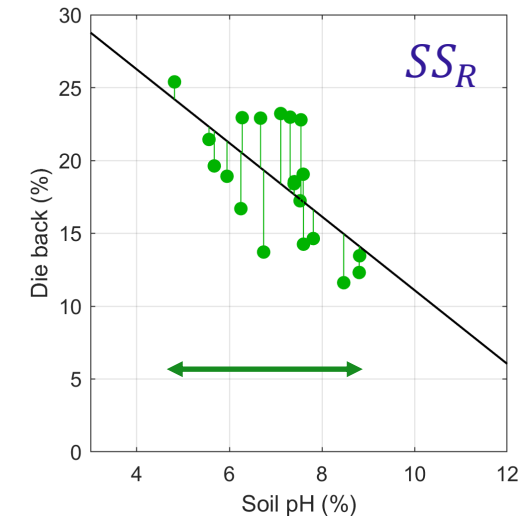
$$CI_1 = \widehat{b}_1 \pm t_{1-\frac{\alpha}{2}, \nu_R} s_{\widehat{b}_1}$$

- With $s_{\widehat{b}_1}$ given by

$$s_{\widehat{b}_1} = \frac{s_\varepsilon}{\sqrt{S_{xx}}}$$

- Proof is beyond the scope of this course
- Considerations
 - Numerator is RMS of residuals on y axis
 - Increases as variation from the model increases
 - Denominator – length scale for the range of the x-axis
 - Decreases as the range of x_i decreases, so larger uncertainty

$$s_\varepsilon = \sqrt{\frac{SS_R}{\nu_R}}$$



$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

Confidence Interval on the Intercept

- The confidence interval will follow the general form

$$CI_0 = \widehat{b}_0 \pm t_{1-\frac{\alpha}{2}, \nu_R} s_{\widehat{b}_0}$$

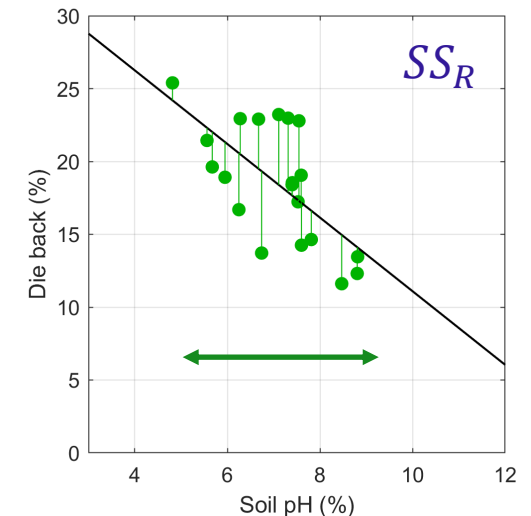
- With $s_{\widehat{b}_0}$ given by

$$s_{\widehat{b}_0} = s_{\varepsilon} \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}$$

- Considerations

- s_{ε} is RMS of residuals
 - Increases as variation from the model increases
- $1/n$ term tells us larger sample sizes are better
- Last term – length scale for the range of the x-axis

$$s_{\varepsilon} = \sqrt{\frac{SS_R}{\nu_R}}$$



$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

95% CI on Slope and Intercept

$S_{xx} = 21.98$

$n = 20$

$CI_0 = \widehat{b}_0 \pm t_{1-\frac{\alpha}{2},v_R} s_{\widehat{b}_0}$

$t_{1-\frac{\alpha}{2},v_R} = 2.101$

$S_{yy} = 329.6$

$\bar{x} = 7.066$

$CI_1 = \widehat{b}_1 \pm t_{1-\frac{\alpha}{2},v_R} s_{\widehat{b}_1}$

$S_{xy} = -55.52$

$\bar{y} = 18.5$

$SS_R = 189.4$

```
>> tinv(0.975,18)
ans =
    2.1009
```

$v_R =$

$s_{\varepsilon} =$

$\widehat{b}_1 =$

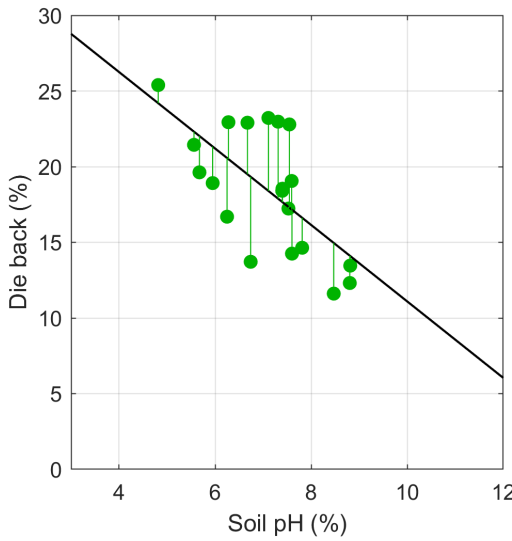
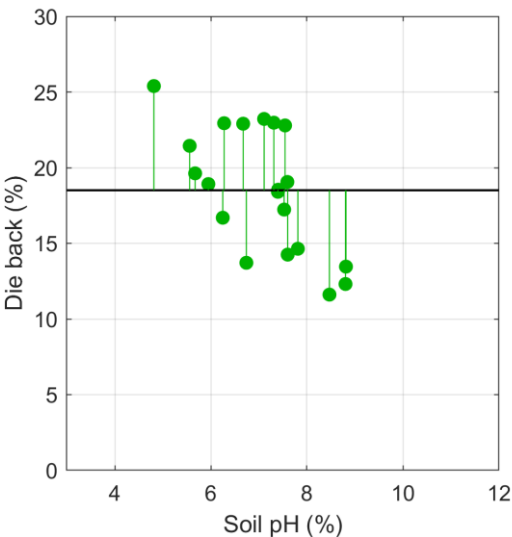
$s_{\widehat{b}_1} =$

$CI_1 =$

$\widehat{b}_0 =$

$s_{\widehat{b}_0} =$

$CI_0 =$



Hypothesis Testing the Slope and Intercept

- $\hat{b}_i$ values are equivalent to mean values
- $s_{\hat{b}_i}$ values are equivalent to standard errors on the fit parameters
- We define our test statistic

$$T_i = \frac{\hat{b}_i}{s_{\hat{b}_i}}$$

- Which follows a t-distribution with ν_R DOF
- Can do one- or two-tailed tests
- E.g. two-tailed
 - H_0 : the slope/intercept is zero, $b_i = 0$
 - H_1 : the slope/intercept is not zero, $b_i \neq 0$
- p -values analysed as per t-tests

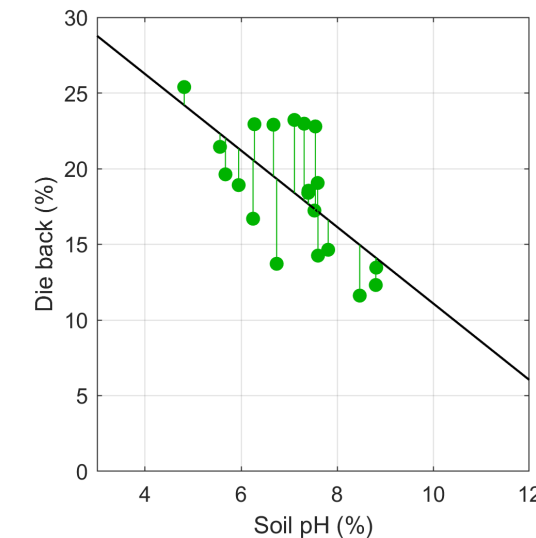
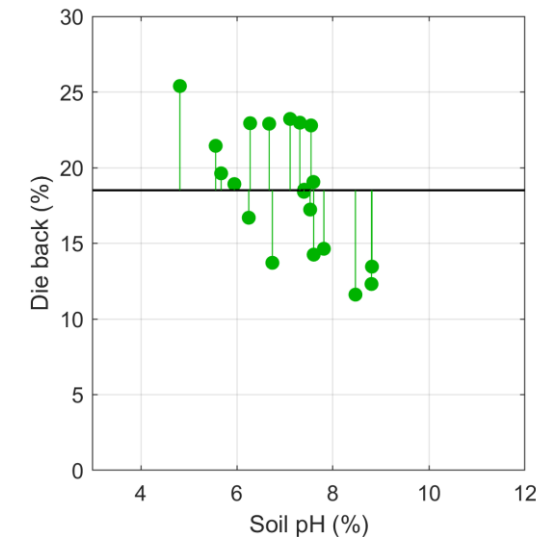
$$b_i > 0$$



$$b_i < 0$$



$$b_i \neq 0$$



Hypothesis Testing Slope and Intercept

$$S_{xx} = 21.98$$

$$n = 20$$

$$CI_0 = \widehat{b}_0 \pm t_{1-\frac{\alpha}{2}, v_R} s_{\widehat{b}_0}$$

$$S_{yy} = 329.6$$

$$\bar{x} = 7.066$$

$$CI_1 = \widehat{b}_1 \pm t_{1-\frac{\alpha}{2}, v_R} s_{\widehat{b}_1}$$

$$v_R = 18$$

$$S_{xy} = -55.52$$

$$\bar{y} = 18.5$$

$$SS_R = 189.4$$

$$\widehat{b}_1 = -2.526$$

$$s_{\widehat{b}_1} = 0.692$$

$$\widehat{b}_0 = 36.35$$

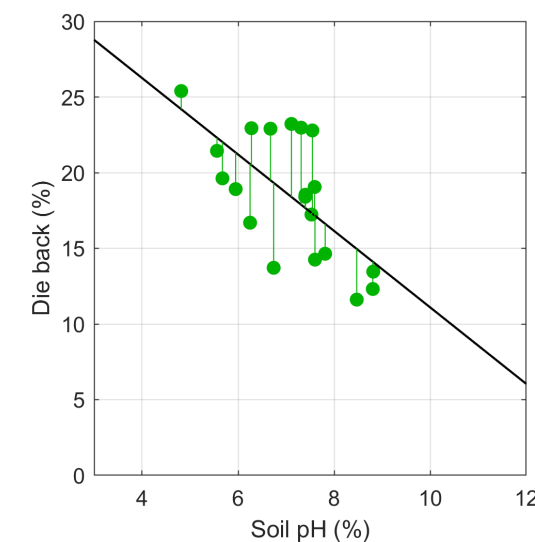
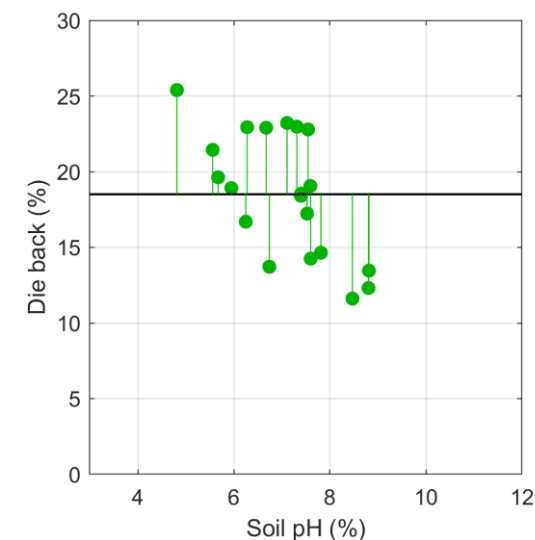
$$s_{\widehat{b}_0} = 4.94$$

$$t_1 = \widehat{b}_1 / s_{\widehat{b}_1}$$

$$t_0 = \widehat{b}_0 / s_{\widehat{b}_0}$$

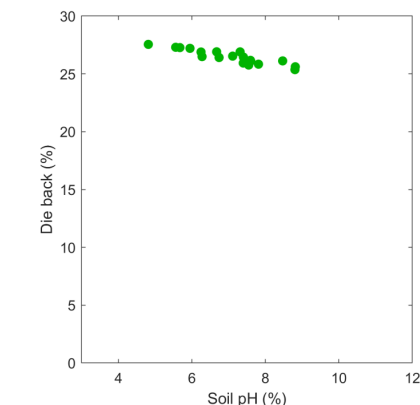
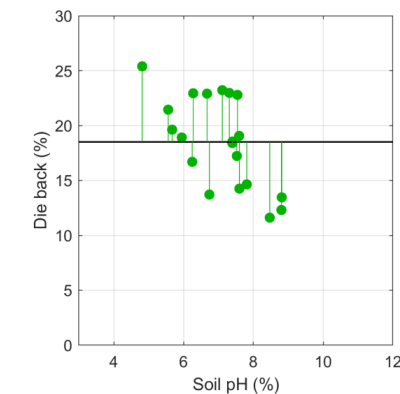
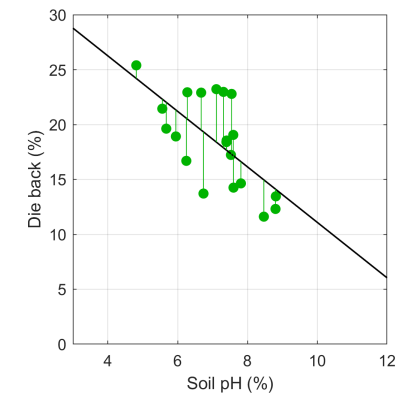
```
>> p1=2*(1-tcdf(abs(t1),v))
p1 =
0.0018
```

```
>> p0=2*(1-tcdf(abs(t0),v))
p0 =
7.9215e-07
```



Implications of H_0 on regression parameters

- If H_0 not rejected for b_1
 - We cannot conclude that $Y = (\widehat{b}_0 + \widehat{b}_1 X)$ is any better than $Y = \widehat{b}_0$ would have been
 - Consider simpler model: $Y = (b_0) + \varepsilon$
 - Note in this context b_0 is the mean
 - Note, H_0 on b_1 is equivalent to H_0 on R
- Consider physical implications of slope
 - Does the slope indicate that X has a practically significant impact on Y ?
- If H_0 not rejected for b_0
 - We cannot conclude that $Y = (\widehat{b}_0 + \widehat{b}_1 X)$ is any better than $Y = \widehat{b}_1 X$ would have been
 - Consider simpler model: $Y = (b_1 X) + \varepsilon$
 - i.e. a straight line forced through zero
- Consider physical implications of intercept
 - Real world example: flow vs pressure in ex vivo eyes
 - When pressure equals zero, no mechanism to drive flow
 - Hence intercept should equal zero
 - Allowing a free parameter means other model parameter (slope) is wrong ([Sherwood et al. 2016](#))





Bands are important

Confidence and Prediction Bands

Confidence Bands

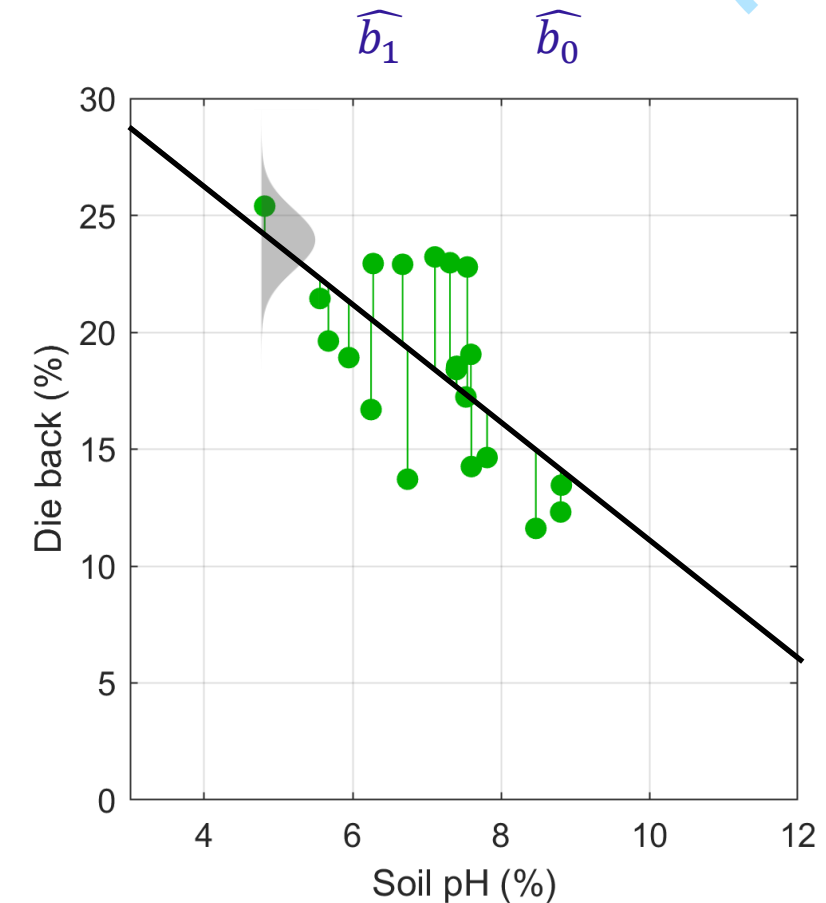
- We calculated confidence intervals on the slope and intercept
- This reminds us that both the slope and intercept are statistics that can vary
- For any given value of x^* we can calculate the expected value $\hat{y}^*$ but also confidence intervals on $\hat{y}^*$

$$CI(x^*) = \hat{y}^*(x^*) \pm t_{1-\frac{\alpha}{2}, \nu_R} s_{\hat{y}^*}(x^*)$$

$$s_{\hat{y}^*}(x^*) = s_{\varepsilon} \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}}$$

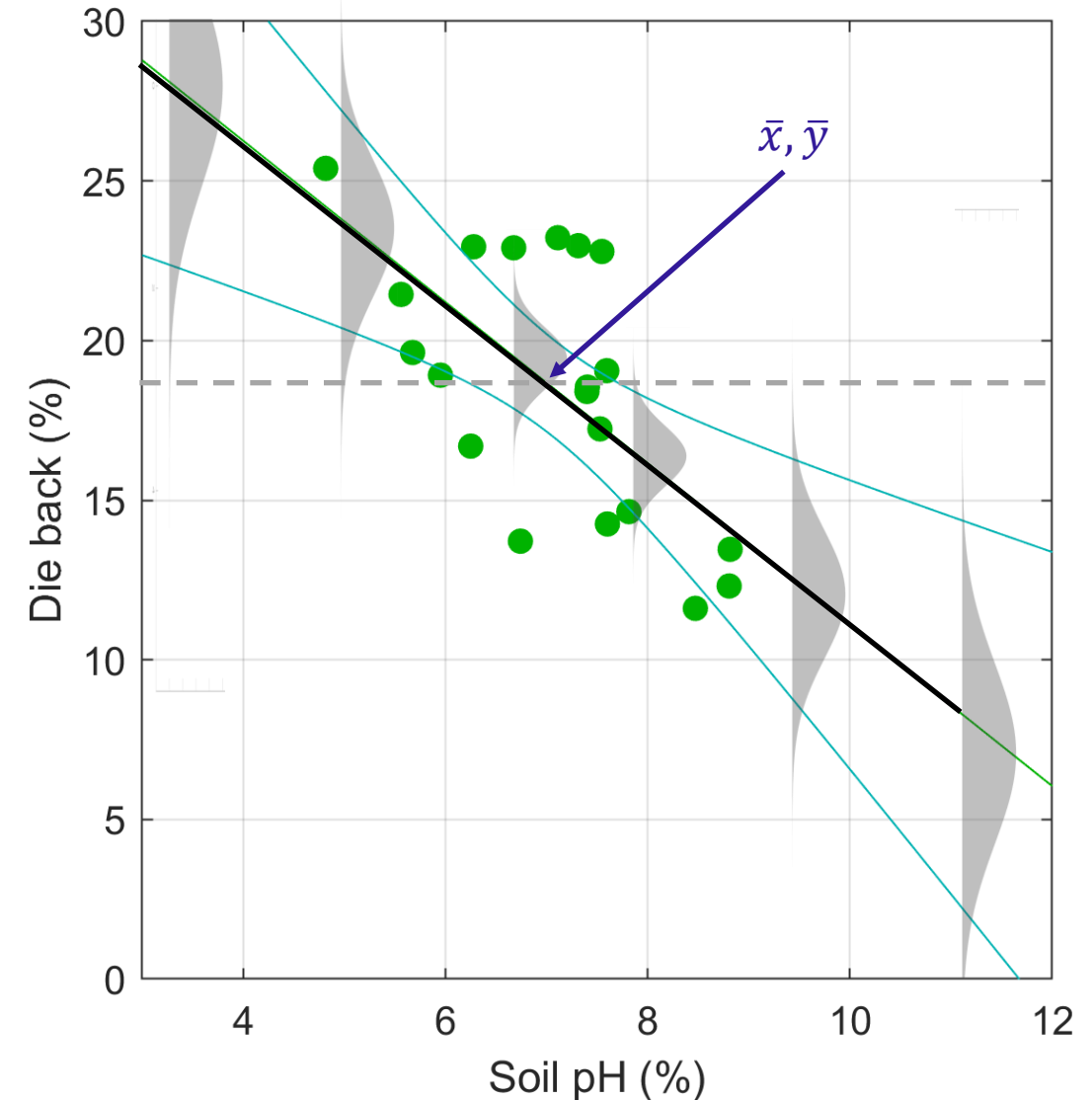
Increases the further we get from $\bar{x}$

Note $s_{\hat{y}^*}(0) = s_{\hat{b}_0}$



Confidence Bands

- 95% confidence bands visualise the 95% confidence interval on the true 'population' line for all x
- For a given x^* , if we took 100 samples and calculated the best linear fit for each, 95 of the CIs on $\hat{y}^*$ would contain the population value of $y(x^*)$
- Instant visualisation of statistical difference from zero
 - if a horizontal line is contained within the confidence bounds, then $p_1 > 0.05$
 - The slope is not statistically different from zero at $\alpha = 0.05$



Prediction Bands

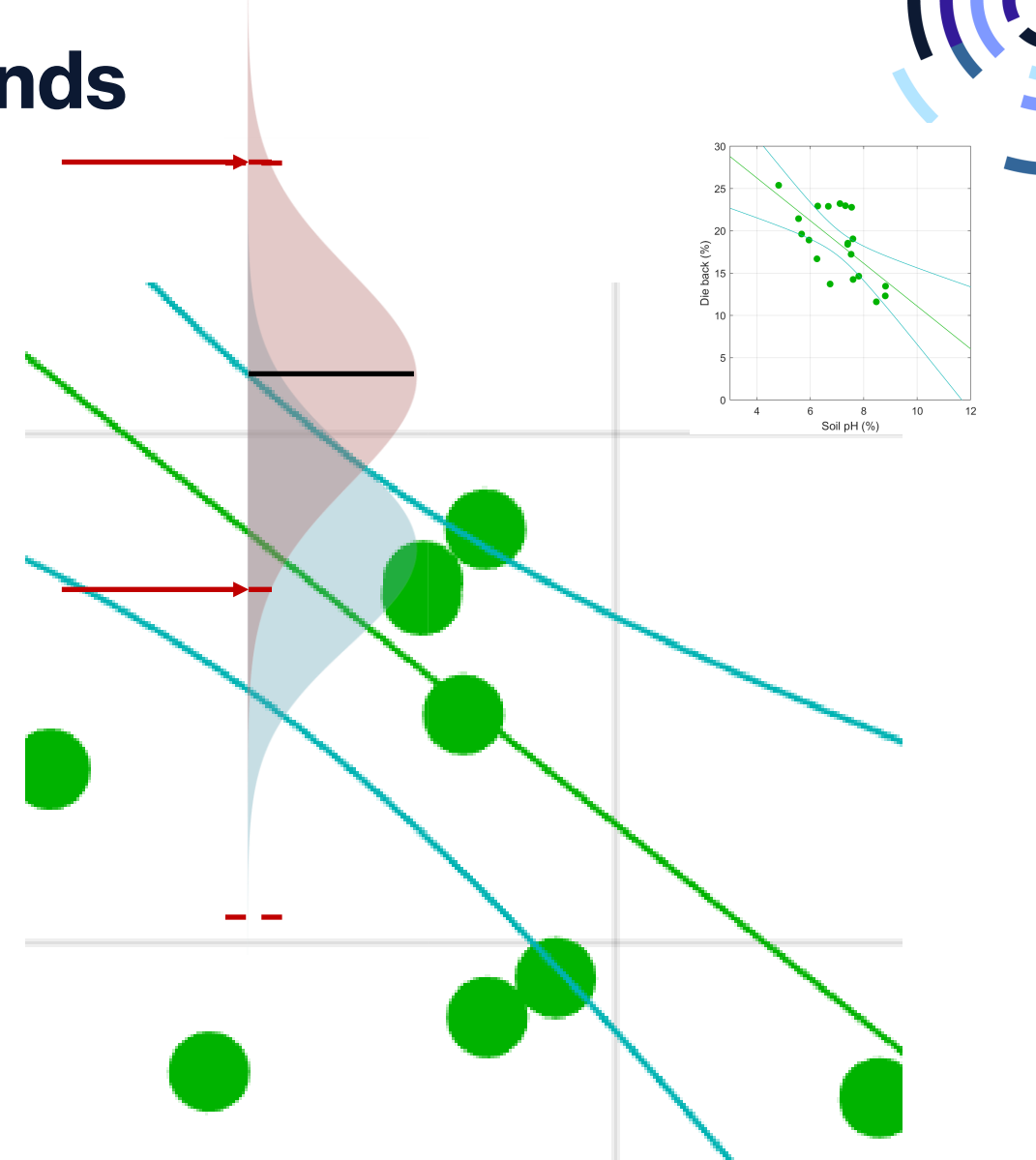
- It is also useful to use our model (a straight line) to predict future events
- As we are considering a single event, rather than just the mean we have two sources of uncertainty to include
 - Uncertainty on the mean (blue)
 - Uncertainty around the mean (variance of y - red)
 - Variances add in quadrature, hence

$$s_{pred}^2(x^*) = s_{\hat{y}^*}^2(x^*) + s_{\varepsilon}^2$$

$$PI(x^*) = \hat{y}^*(x^*) \pm t_{1-\frac{\alpha}{2}, n-2} s_{pred}(x^*)$$

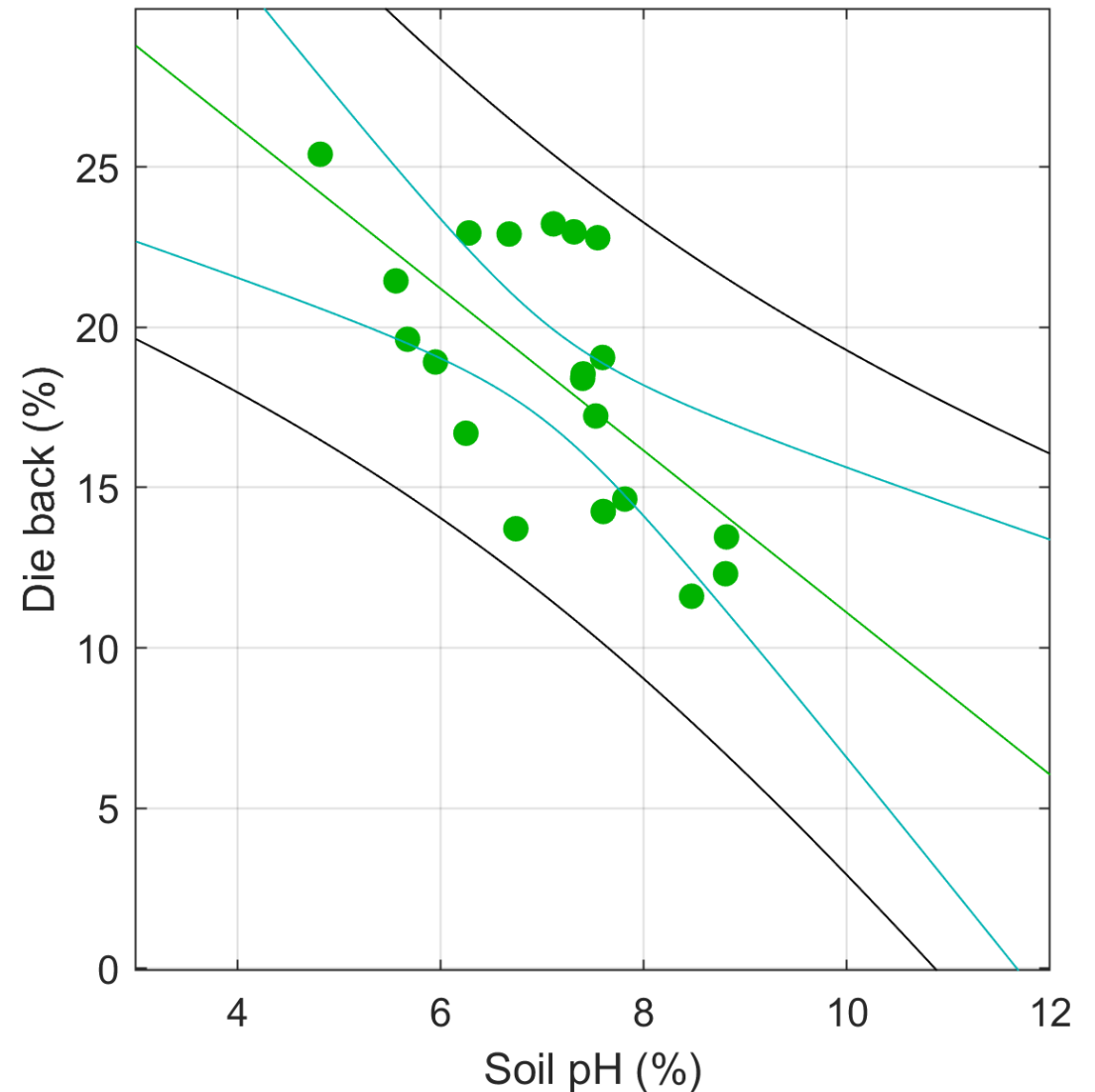
$s_{\hat{y}^*}^2$ from uncertainty on mean

s_{ε}^2 from variability around the mean



Prediction Bands

- 95% prediction bands predict where a future data point will be 95% of the time
- Can be interpreted as – ‘if we took 100 samples, the individual values would lie in this range 95% of the time’
- Confidence bands are analogous to the confidence interval on the mean
 - Important if we only care about the past fit
- Prediction Bands are comparable to prediction intervals
 - Important if we care about a future prediction



Advanced Regression

- Recall our general model formulation

$$outcome_i = (model) + error_i$$

- Thus far we have considered two models
 - The mean
 - A linear fit
- However, data often do not fit into such easy categories
- For these cases, we need a better model
- Common options are
 - A polynomial fit
 - A power law fit
 - A logarithmic or exponential fit
- Non-linear regression can be used to calculate the best fit, confidence intervals, confidence bands and prediction intervals for us for any of these models
- At this point in the course, we just trust the computers to do it for us!
- See bonus rant video for non-linear regression with MATLAB and a case study
- Regression can also be used when there are a mix of continuous and categorical variables or other factors (covariates) that need to be included



Non-parametric Tests and Regression

What we covered

- Non-parametric tests
 - Sign test
 - Sign-rank test
 - Rank-sum test
- Categorical vs continuous analysis
- Correlation Coefficients
 - Pearson's and the Coefficient of Determination
 - Spearman's
- Linear Regression
- Confidence Intervals and NHST on regression parameters
- Confidence and Prediction Bands

Up next

Lecture 9: Multiple Comparisons